

PHYSICS

1) If $\vec{a} = 2\hat{i} + 6\hat{j} + m\hat{k}$ and $\vec{b} = n\hat{i} + 12\hat{j} + 4\hat{k}$ are parallel to each other, then product of m and n is :-

- (1) 2
- (2) 4
- (3) 8
- (4) 16

2) A truck travelling due north at 20 m/s turns west and travels at the same speed. The change in its velocity be :-

- (1) 40 m/s N-W
- (2) $20\sqrt{2}$ m/s N-W
- (3) 40 m/s S-W
- (4) $20\sqrt{2}$ m/s S-W

3) $\int 2 \sin(2x + 3) dx = \dots\dots\dots ?$

- (1) $4 \cos(2x + 3) + C$
- (2) $\frac{\cos(2x + 3)}{2} + C$
- (3) $-\cos(2x + 3) + C$
- (4) $\frac{-\cos(2x + 3)}{4} + C$

4) $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ is equal to-

- (1) $2 \cos\alpha \cos\beta$
- (2) $-2 \cos\alpha \cos\beta$
- (3) $2 \sin\alpha \sin\beta$
- (4) $-2 \sin\alpha \sin\beta$

5) Vector $\vec{A}$ is 2cm long and is 60° above the x-axis in the first quadrant. Vector $\vec{B}$ is 2cm long and is 60° below the x-axis in the fourth quadrant. The sum $\vec{A} + \vec{B}$ is a vector of magnitude

- (1) 2 cm along + y-axis
- (2) 2cm along +x-axis
- (3) 2 cm along -y- axis
- (4) 2 cm along - x- axis

6) If velocity of particle varies with time as $v = (t + 2)(t + 3)$, then its acceleration at $t = 1$ s is :-

- (1) 5 m/s^2
- (2) 7 m/s^2
- (3) 2 m/s^2
- (4) 3 m/s^2

7) The roots of $10x^2 - 27x + 5 = 0$ are :-

- (1) $\frac{2}{5}, \frac{1}{5}$
- (2) $\frac{2}{5}, 5$
- (3) $\frac{5}{2}, \frac{1}{5}$
- (4) $2, \frac{5}{2}$

8) Which of the following group of forces can't produce zero resultant :

- (1) 10, 10, 10
- (2) 10, 10, 20
- (3) 10, 20, 20
- (4) 10, 20, 40

9) If $y = 2x^2 - 2x$ then find out minimum value of y:

- (1) $\frac{1}{2}$
- (2) $-\frac{1}{2}$
- (3) $\frac{3}{2}$
- (4) $-\frac{3}{2}$

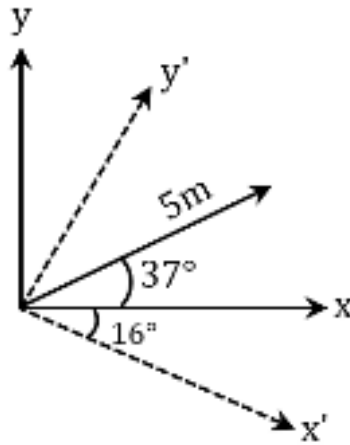
10) **Assertion** : If angle between two vectors $\vec{A}$ and $\vec{B}$ is 45° then $\vec{A} \times \vec{B} = \vec{A} \cdot \vec{B}$.

Reason : At $\theta = 45^\circ$, $\sin \theta = \cos \theta$

Choose correct alternative : -

- (1) Both are correct and reason is correct explanation for assertion.
- (2) Both are correct and reason is not correct explanation for assertion.
- (3) Assertion is wrong, reason is correct.
- (4) Assertion is correct, reason is wrong.

11) A vector of magnitude 5m is placed in a coordinate system as shown in the figure. An auxiliary set of axes (x' and y') is formed by rotating the original axes clockwise by 16° . Find the components



of the vector along the new x' and y' axes.

- (1) (4,3)
- (2) (3,4)
- (3) $\left(\frac{3}{5}, \frac{4}{5}\right)$
- (4) (4,5)

12)

Find the projection of $\hat{i} + 2\hat{j}$ on $2\hat{i} + 3\hat{j}$ in vector form

- (1) $\frac{8}{\sqrt{5}}$
 $(\hat{i} + 2\hat{j})$
- (2) $\frac{8}{13}(2\hat{i} + 3\hat{j})$
- (3) $\frac{6}{\sqrt{13}}(2\hat{i} + 3\hat{j})$
- (4) $\frac{6}{\sqrt{5}}$
 $(\hat{i} + 2\hat{j})$

13) Let $\vec{A}, \vec{B}, \vec{C}$ be three non-zero vectors. Consider the following statements:

- (a) The sum of two vectors does not depend on their order.
- (b) The subtraction of two vectors can be written by reversing the order with a negative sign.
- (c) If the dot product of two vectors is zero, that one of the vectors must be a zero vector.
- (d) The cross product of two vectors changes sign when the order of vectors is reversed.

- (1) only a, b, d are correct
- (2) only a, c, d are correct
- (3) only a, b are correct
- (4) all are correct

14) The main scale of a vernier callipers reads 10 mm in 10 divisions. 10 divisions of Vernier scale coincide with 9 divisions of the main scale. When the two jaws of the callipers touch each other, the fifth division of the vernier coincides with 9 main scale divisions and the zero of the vernier is to the right of zero of main scale. When a cylinder is tightly placed between the two jaws, the zero of

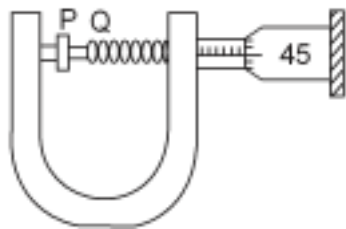
vernier scale lies slightly behind 3.2 cm and the fourth vernier division coincides with a main scale division. The diameter of the cylinder is.

- (1) 3.10 cm
- (2) 3.8 cm
- (3) 3.09 cm
- (4) -3.09 cm

15) If the time of oscillation (T) of a gas bubble from an explosion under water depends on static pressure (P), density of water (d) and total energy of explosion (E), find the relation between them : -

- (1) $T \propto P^{-5/6} d^{1/2} E^{1/3}$
- (2) $T \propto P^{1/2} d^{1/3} E^{1/2}$
- (3) $T \propto P^{-5/6} d^{1/3} E^{1/2}$
- (4) $T \propto P^{-2/3} d^{1/2} E^{1/3}$

16) In an experiment to find out the diameter of wire using screw gauge, the following observations



were noted :

(a) Screw moves 0.5 mm on main scale in one complete rotation

- (b) Total divisions on circular scale = 50
- (c) Main scale reading is 2.5 mm
- (d) 45th division of circular scale is in the pitch line
- (e) Instrument has 0.03 mm negative error

Then the diameter of wire is :

- (1) 2.92 mm
- (2) 2.54 mm
- (3) 2.98 mm
- (4) 3.45 mm

17) Which of the following relation is dimensionally correct for the terminal velocity (V) of a rain drop? (where r is radius, g is acceleration due to gravity, ρ is density of drop, σ is density of air and η is viscosity) : -

- (1) $V = \frac{2}{9} \frac{r^2 g (\rho - \sigma)}{\eta}$
- (2) $V = \frac{2}{9} \frac{r g (\rho - \sigma)}{\eta}$
- (3) $V = \frac{2}{9} \frac{r^2 \eta (\rho - \sigma)}{g}$
- (4) $V = \frac{2}{9} \frac{\eta g}{r^2 (\rho - \sigma)}$

18) **Assertion** : Light year and parallax second used to measure time.

Reason : Both light year and parallax second have the dimensions of length (L).

- (1) Both A & R are true, R is correct explanation of A.
- (2) Both A & R are true, R is not correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

19) The mass and volume of a body are found to be (5.00 ± 0.05) kg and (1.00 ± 0.05) m³ respectively. Then the maximum possible percentage error in density :-

- (1) 3%
- (2) 10%
- (3) 6%
- (4) 5%

20) In each of the following expression A, B and C represent lengths :-

- I. $A + B$ II. AB
III. $\frac{AB}{C}$ IV. $\frac{A}{(B + C)}$

Which of the following statements is incorrect ?

- (1) One of them has dimensions of volume
- (2) One of them has dimensions of area
- (3) Two of them have dimensions of length
- (4) One of them is dimensionless

21) The frequency of vibration f of a mass m suspended from a spring of spring constant k is given by a relation of this type $f = cm^x k^y$; where c is dimensionless quantity. The values of x and y are :-

- (1) $x = \frac{1}{2}, y = \frac{1}{2}$
- (2) $x = \frac{-1}{2}, y = \frac{-1}{2}$
- (3) $x = \frac{1}{2}, y = \frac{-1}{2}$
- (4) $x = \frac{-1}{2}, y = \frac{1}{2}$

22) In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and unit of length is 10m. In this system, one unit of power will correspond to

- (1) 16 watts
- (2) 1/16 watts

- (3) 25 watts
(4) none of these

23) Match the physical quantities given in column I with dimensions in column II :-

Column I		Column II	
a	Torque	P	ML^2T^{-1}
b	Angular momentum	q	$[M^0L^2T^{-2}]$
c	Latent heat	r	$M^0L^0T^1$
d	Surface tension	s	M^0LT^0
e	Light year	t	ML^2T^{-2}
		u	MLT^{-2}

- (1) a - t, b - p, c - q, d - u, e - s
(2) a - q, b - r, c - p, d - t, e - s
(3) a - q, b - p, c - t, d - r, e - s
(4) a - t, b - s, c - q, d - p, e - r

24)

For $z = a^2x^3y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12%, respectively, then the percentage error for 'z' will be %.

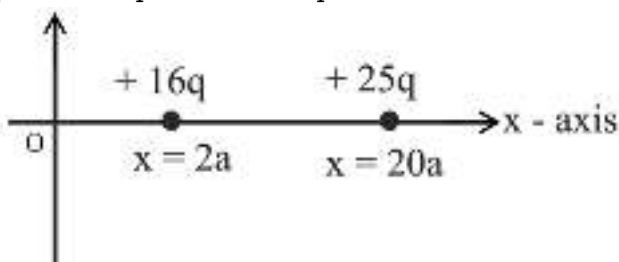
- (1) 14%
(2) 18%
(3) 20%
(4) 22%

25) **Assertion (A) :** - The number of significant figures in 0.005 is one, while in 0.500 it is three.

Reason (R) : - Zeros to the left of the first non-zero digit are not significant, but trailing zeros in a number with a decimal points are significant.

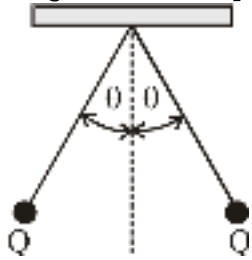
- (1) A & R are correct & R is correct explanation of A.
(2) A & R are correct but R is not correct explanation of A.
(3) A is correct but R is not correct.
(4) A is not correct but R is correct.

26) Two charge particles are shown in the diagram. Where should a third charge particle q_0 be placed so q_0 will be in equilibrium :-



- (1) $x = 10 a$
- (2) $x = 8 a$
- (3) $x = 11 a$
- (4) $x = 12 a$

27) Two pith balls each with mass m are suspended from insulating threads. When the pith balls are given equal positive charge Q , they hang in equilibrium as shown. We now increase the charge on the left pith ball from Q to $2Q$ while leaving its mass essentially unchanged. Which of the following diagrams best represents the new equilibrium configuration?



- (1)
- (2)
- (3)
- (4)

28) A charge particle q_1 is at position $(2, -1, 3)$. The electrostatic force on another charged particle q_2 at $(0, 0, 0)$ is :

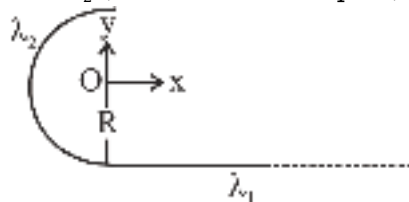
- (1) $\frac{q_1 q_2}{56\pi\epsilon_0} (2\hat{i} - \hat{j} + 3\hat{k})$
- (2) $\frac{q_1 q_2}{56\sqrt{14}\pi\epsilon_0} (2\hat{i} - \hat{j} + 3\hat{k})$
- (3) $\frac{q_1 q_2}{56\pi\epsilon_0} (\hat{j} - 2\hat{i} - 3\hat{k})$
- (4) $\frac{q_1 q_2}{56\sqrt{14}\pi\epsilon_0} (\hat{j} - 2\hat{i} - 3\hat{k})$

29) Two negative charge, each of magnitude q are situated at $2r$ distance apart. A positive charge q

is lying at the centre between them. The potential energy of the system is U_1 . If the two nearest charges are mutually interchanged and the potential energy becomes U_2 , then $\frac{U_1}{U_2}$ will be :-

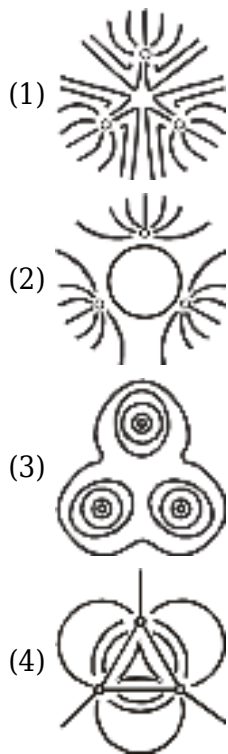
- (1) 3
- (2) 5
- (3) 2
- (4) 1

30) In the figure shown, find the ratio of the linear charge densities λ_1 (on semi-infinite straight wire) and λ_2 (on semi-circular part) that is λ_1/λ_2 so that the field at O is along y direction.



- (1) 2
- (2) 1.5
- (3) 3
- (4) 2.5

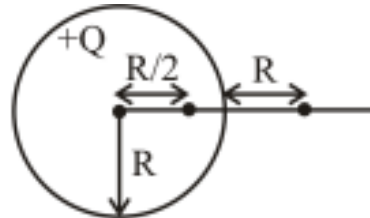
31) Three positive charges of equal value q are placed at vertices of an equilateral triangle. The resulting lines of force should be sketched as in :-



32) Two identical spheres carrying charges $-9\mu\text{C}$ and $5\mu\text{C}$ respectively are kept in contact and then separated from each other. Point out true statement from the following. On each sphere (after contact) :-

- (1) 1.25×10^{13} electrons are in deficit
- (2) 1.25×10^{13} electrons are in excess
- (3) 2.15×10^{13} electrons are in excess
- (4) 2.15×10^{13} electrons are in deficit

33) A point charge (+q) of mass m is released from a distance $\left(\frac{R}{2}\right)$ from the centre of a solid sphere of radius R and charge Q (uniformly charged in volume). The speed of the point charge when it reaches a distance R from the surface of the sphere (there is no interaction between sphere and



point charge other than electrostatic). $k = \frac{1}{4\pi\epsilon_0}$ fixed solid sphere

- (1) $\sqrt{\frac{7kQq}{16mR}}$
- (2) $\sqrt{\frac{7kQq}{4mR}}$
- (3) $\sqrt{\frac{5kQq}{16mR}}$
- (4) $\sqrt{\frac{3kQq}{4mR}}$

34)

Electric potential at a point is: $V = x^2y + yz$. The electric field at the point (1, 3, 1) is

- (1) 7 units
- (2) 70 units
- (3) 49 units
- (4) 490 units

35) An electric dipole of moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. Angle between $\vec{p}$ and $\vec{E}$ is 30° . Calculate work done by field when angle is increased to 60° .

- (1) $\frac{pE}{2}(\sqrt{3} - 1)$
- (2) $\frac{pE}{2}(1 - \sqrt{3})$
- (3) $\frac{pE}{(\sqrt{3} - \sqrt{2})}$
- (4) zero

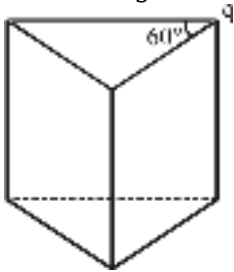
36) The figure shows a charge q placed inside a cavity in an uncharged conductor. Now if an

external electric field is switched on then :



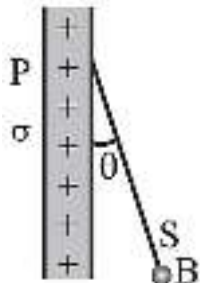
- (1) only induced charge on outer surface will redistribute
- (2) only induced charge on outer surface will redistribute
- (3) Both induced charge on outer and inner surface will redistribute
- (4) force on charge q placed inside the cavity will change

37) A charge q is placed at a vertex of the given geometry of angle 60° . Then, the flux coming out from this geometry will be



- (1) $\frac{q}{6\epsilon_0}$
- (2) $\frac{q}{16\epsilon_0}$
- (3) $\frac{q}{8\epsilon_0}$
- (4) $\frac{q}{12\epsilon_0}$

38) A charged ball B hangs from a silk thread S, which makes an angle θ with a large charged conducting sheet P, as shown in the figure. The surface charge density σ of the sheet is proportional

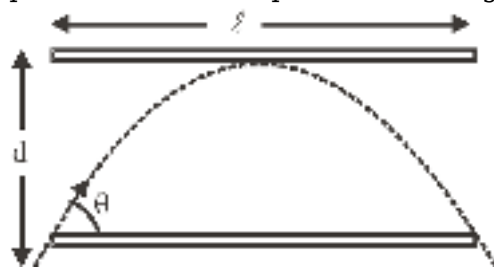


to :-

- (1) $\sin \theta$
- (2) $\cos \theta$
- (3) $\tan \theta$
- (4) $\cot \theta$

39) An electron enter the region between the plates of a parallel plate as shown in figure. The plate

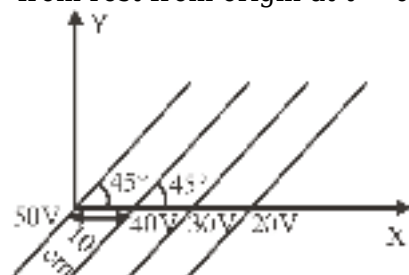
length is ℓ & separation is d . Electric field between plates is uniform. The electron misses the upper plate & follows the path shown. Neglect gravity.



then correct relation is :-

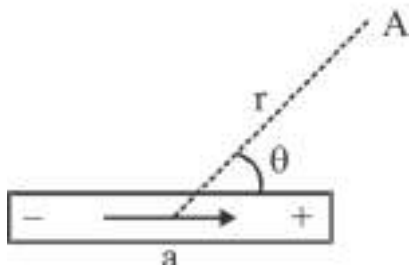
- (1) $\tan \theta = 2d / \ell$
- (2) $\tan \theta = 4d / \ell$
- (3) $\tan \theta = 8d / \ell$
- (4) The data given is insufficient to find a relation among d, ℓ, θ

40) In the figure equipotential surfaces are shown. A particle of charge q and mass m is released from rest from origin at $t = 0$. position vector of particle at $t = 1$ sec is :-



- (1) $100 \frac{q}{m} (\hat{i} - \hat{j})m$
- (2) $50 \frac{q}{m} (\hat{i} - \hat{j})m$
- (3) $50 \frac{q}{m} (-\hat{i} + \hat{j})m$
- (4) $100 \frac{q}{m} (-\hat{i} + \hat{j})m$

41) Electric field at a distance r ($r \gg a$) is E_1 , when angle θ is 30° . If only θ changes to 45° then



electric field is :-

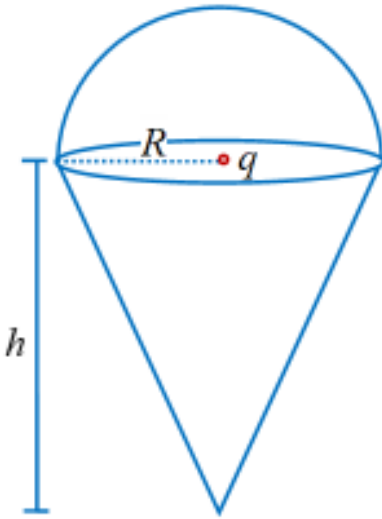
- (1) $\sqrt{\frac{10}{7}} E_1$
- (2) $\sqrt{\frac{7}{10}} E_1$

(3) $\sqrt{\frac{10}{13}}E_1$

(4) E_1

42) A charge q is surrounded by a closed surface consisting of an inverted cone of height h and base radius R , and a hemisphere of radius R as shown in the figure. The electric flux through the conical

surface is $\frac{nq}{6\epsilon_0}$ (in SI units). The value of n is _____.



(1) 3

(2) 6

(3) 1

(4) 2

43) A dipole $\vec{p} = p\hat{i} + p\hat{j}$ is placed in an electric field $\vec{E} = (ax)\hat{i} + (by)\hat{j}$. The net force on the dipole is:

(1) $p(a + b) (\hat{i} + \hat{j})$

(2) $pa\hat{i} + pb\hat{j}$

(3) $\frac{(pa + pb)}{K}$

(4) Zero

44) A point charge of 10^{-8} C is placed at origin. The work done in moving a point charge $2 \mu\text{C}$ from point A(4, 4, 2) m to B(2, 2, 1) m is _____J.

$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ in SI units)

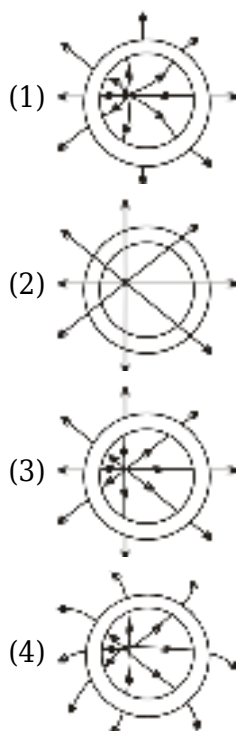
(1) 45×10^{-6}

(2) 0

(3) 30×10^{-6}

(4) 15×10^{-6}

45) A point charge q is placed at a point inside a hollow conducting sphere. Which of the following electric force pattern is correct ?



CHEMISTRY

1) In the hydrogen atoms, the electrons are excited to the 5th energy level. The number of the lines that may appear in the emission spectrum will be :

- (1) 4
- (2) 8
- (3) 10
- (4) 12

2) If radius of second stationary orbit (in Bohr's atom) is R . Then radius of third orbit will be:-

- (1) $R/3$
- (2) $9R$
- (3) $R/9$
- (4) $2.25 R$

3) How much energy must be provided to an electron present in ground state of hydrogen atom, so that electron excites to 2nd excited state.

- (1) 10.2 eV
- (2) 13.6 eV
- (3) 1.89 eV
- (4) 12.09 eV

4) ${}^{11}_5\text{B}$ and ${}^{13}_6\text{C}$ are related to each other as :-

- (1) Isotope
- (2) Isodiapher
- (3) Isosters
- (4) Isoelectronic

5) Which species are isoelectronic

(NO_3^- , CO_3^{-2} , ClO_3^- , SO_3)

- (1) NO_3^- , CO_3^{-2}
- (2) SO_3 , NO_3^-
- (3) ClO_3^- , CO_3^{-2}
- (4) CO_3^{-2} , SO_3

6) X^{+2} has 10 protons then find number of protons in X^{-2} :

- (1) 10
- (2) 12
- (3) 14
- (4) 16

7) In ${}_{92}\text{U}^{235}$ nucleus, the number of electrons is :-

- (1) 92
- (2) 235
- (3) 143
- (4) Zero

8) Ratio of mass of neutron and proton is :-

- (1) Less than one
- (2) More than one
- (3) Exactly equal to one
- (4) Can't be predicted

9) Rutherford's experiment on scattering of alpha particles showed for the first time that atom has:

- (1) Electrons
- (2) Protons
- (3) Nucleus
- (4) Neutrons

10) Which of the following conclusions regarding the structure of atom is true based on Rutherford's α -particle scattering experiment ?

- (1) The positive charge is concentrated in a very small volume of the atom
- (2) The positive charge is scattered with the electrons throughout the atom
- (3) The volume occupied by the nucleus is half of the volume of atom
- (4) Most of the space in the atom is occupied by the neutrons.

11) Angular momentum for P-shell electron is :-

- (1) $\frac{3h}{\pi}$
- (2) Zero
- (3) $\frac{\sqrt{2}h}{2\pi}$
- (4) None

12) The mass no. of an element is 39. The number of neutrons in its nucleus is one more than the number of protons. The number of protons, neutrons and electrons respectively in its atom would be

- (1) 19, 20, 19
- (2) 19, 19, 20
- (3) 20, 19, 19
- (4) 20, 19, 20

13) Energy of e^- in 2nd orbit of H-atom is-

- (1) 13.6 eV
- (2) -3.4 eV
- (3) -1.51 eV
- (4) -0.54 eV

14) If the K.E. of e^- in first and second orbit of Li^{+2} ion is x and y respectively then :

- (1) $x < y$
- (2) $x > y$
- (3) $x = y$
- (4) None of these

15) Find out ratio of following

$(v_{\max})_{\text{Lyman}} : (v_{\max})_{\text{Brackett}}$

- (1) 1 : 16
- (2) 16 : 1
- (3) 4 : 1
- (4) 1 : 4

16) Moles of O_3 in its 12.044×10^{22} molecules of O_3 will be :

- (1) 2 mol
- (2) 0.4 mol
- (3) 1 mol
- (4) 0.2 mol

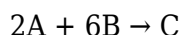
17) If 5 moles of A is mixed 15 moles of B then what is the mole fraction of A in the mixture ?

- (1) 0.75
- (2) 0.35
- (3) 0.25
- (4) 0.5

18) Volume of $\text{H}_2(\text{g})$ required to react completely with 3L of $\text{N}_2(\text{g})$ to form ammonia at same T & P condition will be :

- (1) 7 L
- (2) 6 L
- (3) 9 L
- (4) 4 L

19) When 5 moles of A will react with 3 moles of B according to given balanced reaction



Then which of the following reactant will be in excess amount

- (1) A
- (2) B
- (3) can't be determined
- (4) Both will act as limiting reagent

20) % of H in methane will be :

- (1) 25%
- (2) 30%
- (3) 17%
- (4) 32%

21) The no. of atoms of A and B in their molecule are 9×10^{20} and 3×10^{20} respectively then their empirical formula will be :

- (1) AB_3
- (2) A_3B
- (3) AB
- (4) A_9B

22) 2 moles of urea is dissolved in 20 kg of solvent then molality of the solution formed is :

- (1) 0.1 m
- (2) 0.2 m
- (3) 1 m
- (4) 1.5 m

23) The ratio of the energy of a photon of wavelength 3000 Å to that of a photon of wavelength 6000 Å is :

- (1) $\frac{1}{2}$
- (2) 2
- (3) 3
- (4) $\frac{1}{3}$

24) According to Bohr's theory, the angular momentum to an electron in 5th orbit is :

- (1) 25 (h/π)
- (2) 1.0 (h/π)
- (3) 10 (h/π)
- (4) 2.5 (h/π)

25) When electrons are de-exciting from nth orbit of hydrogen atoms to ground state, 15 spectral lines are formed. The shortest wavelength among these will be :

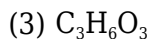
- (1) $\frac{11}{900}R$
- (2) $\frac{36}{35}R$
- (3) $\frac{35}{36}R$
- (4) $\frac{35}{36}R$

26) A gaseous mixture contains CO₂(g) and N₂O(g) in 2:5 ratio by mass. The ratio of the number of molecules of CO₂(g) and N₂O (g) is:

- (1) 5 : 2
- (2) 2 : 5
- (3) 1 : 2
- (4) 5 : 4

27) The empirical formula of a compound of molecular mass 120 is CH₂O. The molecular formula of the compound is:

- (1) C₂H₄O₂
- (2) C₄H₈O₄



(4) All of these

28) If the velocity of an electron in 1st orbit of H-atom is u , what will be the velocity of electron in 3rd orbit Li^{+2} ?

(1) u

(2) $\frac{u}{3}$

(3) $3u$

(4) $9u$

29) Vapour density of any gas is :-

(1) Unitless quantity

(2) $\frac{\text{Molar mass}}{2}$

(3) Density of any gas with respect to density of H_2 gas under same condition of P and T

(4) All of the above

30) A compound contains 11.5% by mass of sodium. Its minimum molecular weight will be :-

(1) 80 amu

(2) 200 amu

(3) 250 amu

(4) 300 amu

31)

What will be the uncertainty in velocity of an electron if uncertainty in its position is $1\text{\AA}$

(1) $8.76 \times 10^6 \text{ m/s}$

(2) $5.76 \times 10^5 \text{ m/s}$

(3) $5 \times 10^{-24} \text{ m/s}$

(4) $1 \times 10^{23} \text{ m/s}$

32) The total number of atoms present in 1 mol of Ammonium dichromate $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ is :-

(1) 19

(2) N_A

(3) 114.437×10^{23}

(4) 84.322×10^{23}

33) The difference in angular momentum associated with the electron in 3rd and 1st orbits of hydrogen atom is :-

(1) $\frac{h}{\pi}$

(2) $\frac{h}{2\pi}$

(3) $\frac{h}{2}$

(4) $(n-1)\frac{h}{2\pi}$

34) The ratio of the radius of two bohr orbits of Li^{+2} is 1 : 9, what would be there nomenclature :-

(1) K & L

(2) L & M

(3) K & M

(4) K & N

35) A gas absorbs a photon of 300 nm and emits out two wavelengths. If one of the emissions is at 600 nm, the other is at :-

(1) 1035 nm

(2) 1200 nm

(3) 600 nm

(4) 518 nm

36) Which of the following is correct sequence of specific charge ?

(1) $e^- > \text{Na}^+ > p > n$

(2) $e^- > p > \text{Na}^+ > n$

(3) $\text{Na}^+ > e^- > p > n$

(4) $e^- > n > p > \text{Na}^+$

37) According to Bohr's theory, the angular momentum for an electron in its 3rd excited state will be :

(1) $\frac{4h}{3\pi}$

(2) $\frac{2h}{\pi}$

(3) $\frac{5h}{\pi}$

(4) $\frac{25h}{\pi}$

38) The ratio of velocity of the electron in the third and fifth orbit of Li^{2+} would be :-

(1) 3 : 5

(2) 5 : 3

(3) 25 : 9

(4) 9 : 25

39) If the ionization energy for He^+ ion's e^- is x then the energy of e^- in its 1st orbit will be

(1) $-x/2$

(2) $13x/2$

(3) $-x$

(4) $-x/3$

40) The first emission line in H-atom spectrum in balmer series will have wave no. :-

(1) $\frac{5R}{36}$

(2) $\frac{3R}{4}$

(3) $\frac{7R}{4}$

(4) $\frac{9R}{400}$

41) No. of moles of e^- in 12g C-atom will be :

(1) 6 moles

(2) 3 moles

(3) 12 moles

(4) 4 moles

42) Moles of atom in 32 g oxygen gas is :

(1) 2 mol

(2) 3 mol

(3) 1 mol

(4) None of these

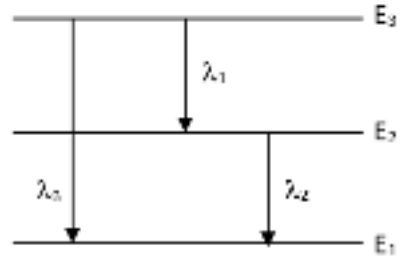
43) In complete combustion of 3g ethane what is the volume of CO_2 gas produced at STP

(1) 44.8 L

(2) 2.24 L

(3) 4.48 L

(4) 11.2 L



44) In the following transition which statement is **correct**?

- (1) $E_{3-1} = E_{3-2} - E_{2-1}$
- (2) $\lambda_3 = \lambda_1 + \lambda_2$
- (3) $\nu_3 = \nu_1 + \nu_2$
- (4) All of these

45)

The ratio of Ionisation energies of Li^{+2} and Be^{+3} is

- (1) $\frac{2}{3}$
- (2) $\frac{3}{2}$
- (3) $\frac{9}{16}$
- (4) $\frac{16}{9}$

BIOLOGY



1) The given animal belongs to which phylum?

- (1) Platyhelminthes
- (2) Aschelminthes
- (3) Annelida
- (4) Arthropoda

2) Match the following column I and II to select correct answer.

Snakes	Common name
1. <i>Vipera</i>	a. Wall lizard

2. <i>Naja</i>	b. Ajar
3. <i>Hemidactylus</i>	c. Indian cobra
4. <i>Python</i>	d. Viper snake

- (1) 1-c 2-d 3-a 4-b
(2) 1-a 2-b 3-c 4-d
(3) 1-d 2-c 3-a 4-b
(4) 1-d 2-c 3-b 4-a

3) Which of the following group of animals exhibit bilateral symmetry only ?

- (1) *Pennatula, Adamsia, Pleurobrachia*
(2) *Euspongia, Spongilla, Taenia*
(3) *Ascaris, Nereis, Scorpion*
(4) *Echinus, Gorgonia, Sycon*

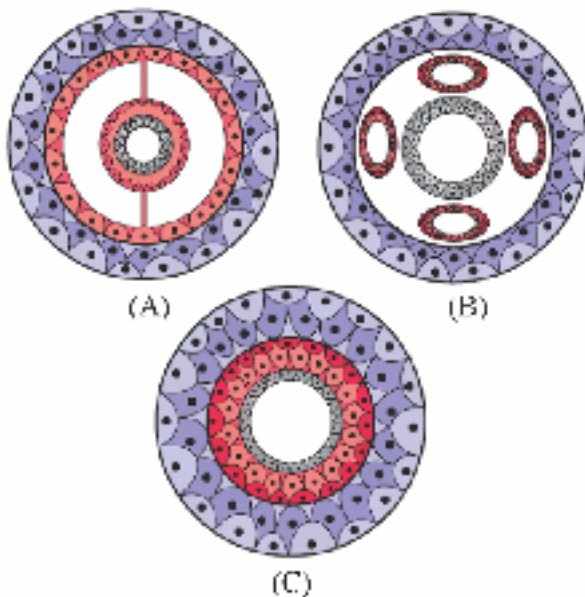
4) Read the following statements from A-D ?

- (A) They have pneumatic bones
(B) Skin is dry without glands.
(C) Digestive tract has additional chambers, the crop and gizzard.
(D) They are warm blooded.

These statements are correct for which animal ?

- (1) *Vipera*
(2) *Columba*
(3) *Labeo*
(4) *Pteropus*

5) In the given diagram identify the coelomic condition in the animals :-



- (1) (A) Tapeworm, (B) Ascaris, (C) Earthworm
(2) (A) Cockroach, (B) Ascaris, (C) Tapeworm
(3) (A) Ascaris, (B) Tapeworm, (C) Earthworm

(4) (A) Earthworm, (B) Cockroach, (C) Ascaris

6) Match the columns I, II, III and choose the correct combination from the given options :-

Column-I	Column-II	Column-III
a. 	1. <i>Aurelia</i>	R. Polyp
b. 	2. <i>Adamsia</i>	S. Medusa

(1) a - 1 - R, b - 2 - S

(2) a - 1 - S, b - 2 - R

(3) a - 2 - R, b - 1 - S

(4) a - 2 - S, b - 1 - R

7) The character given below are shown by :

(i) Extracellular and intracellular digestion.

(ii) Exclusively marine, radially symmetrical, diploblastic, tissue level of organisation.

(iii) Bisexual, fertilization external and indirect development.

(iv) Only sexual reproduction.

(1) Cnidaria

(2) Porifera

(3) Ctenophora

(4) Platyhelminthes

8) Select **incorrect** statement about flatworms :-

(1) Hooks and suckers are present in all flat worms.

(2) Fertilization is internal and development occurs through many larval stages.

(3) Some of them absorb nutrients through body surface.

(4) Locomotory organs are absent.

9) Find out incorrect statement :-

(1) Body of molluscs is segmented as it is divided into head, muscular foot and visceral hump.

(2) Gills help in excretion in Molluscs.

(3) Anus of Mollusc open into mantle cavity

(4) *Loligo* is also known as squid

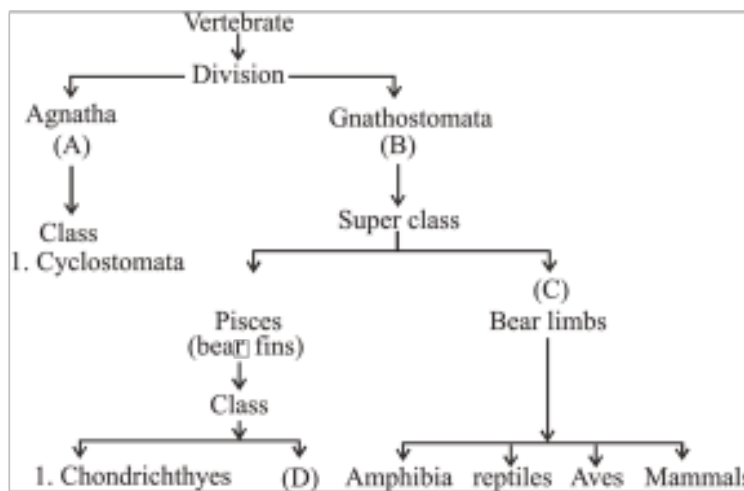
10) How many animals are related with echinodermata ?

Asterias, Echinus, Sea lily, Ophiura, Cucumaria, Saccoglossus, Cuttle fish, Sea hare, Culex, Chiton

- (1) 6
- (2) 4
- (3) 7
- (4) 5

11) Proboscis gland is excretory in function, it is found in :

- (1) *Echinus*
- (2) *Saccoglossus*
- (3) *Balanoglossus*
- (4) Both 2 and 3



12) and identify A, B, C and D :-

Read the following chart given above

	A	B	C	D
(1)	Lacks Jaw	Bears Jaw	Apoda	Osteichthyes
(2)	Bears Jaw	Lacks Jaw	Octapoda	Osteichthyes
(3)	Lacks Jaw	Lacks Jaw	Hexapoda	Osteichthyes
(4)	Lacks Jaw	Bears Jaw	Tetrapoda	Osteichthyes

- (1) 1
- (2) 2
- (3) 3
- (4) 4



13) Excretory and respiratory organs of the given animal are repectively :-

- (1) Malpighian tubules and Book lungs

- (2) Antennary glands and Book lungs
- (3) Coxal glands and Book lungs
- (4) Coxal glands and Trachea

14) Name the first phylum in which alimentary canal is complete with a well developed muscular pharynx -

- (1) Platyhelminthes
- (2) Aschelminthes
- (3) Ctenophora
- (4) Coelenterata

15) See following specimens and give correct answer about them :-



How many of the following features are not common in

both ?

- | | |
|------------------|------------------------|
| (A) Marine | (B) Air bladder |
| (C) Cold blooded | (D) Terminal mouth |
| (E) Oviparous | (F) Presence of scales |

- (1) Two
- (2) Three
- (3) Four
- (4) Five

16) Protochordates are :-

- (1) Non chordates
- (2) Euchordates
- (3) Tunicates and cephalochordates
- (4) Hemichordates

17) Select the correct option and complete the analogy: Egg laying mammal: Platypus :: Flying mammal : ____.

- (1) *Corvus*
- (2) *Macropus*
- (3) *Exocoetus*
- (4) *Pteropus*

18) Which is not an example of mammal ?

- (1) *Macropus*
- (2) *Struthio*
- (3) *Pteropus*
- (4) Both 2 & 3

19) In which one of the following the genus name, its two characters and its class/phylum are correctly matched ?

	Genus Name	Two characters		Class/Phylum
(1)	<i>Aurelia</i>	(a)	Cnidoblasts	Coelentrata
		(b)	Organ level of organization	
(2)	<i>Ascaris</i>	(a)	Body segmented	Annelida
		(b)	Males and females distinct	
(3)	<i>Salamandra</i>	(a)	A tympanum represents ear	Amphibia
		(b)	Fertilization is external	
(4)	<i>Pteropus</i>	(a)	Skin possesses hair	Mammalia
		(b)	Oviparous	

- (1) 1
- (2) 2
- (3) 3
- (4) 4

20)

Read the following four statements A-D :-

- (A) Amphibians live in terrestrial habitats only.
- (B) In amphibians, respiration takes place by gills, lungs, moist skin and through buccopharyngeal cavity.
- (C) The skin of amphibians is smooth, moist and scaleless.
- (D) In Amphibians, Alimentary canal, urinary tract and genital ducts open into a common chamber called cloaca.

How many of the above statements are incorrect?

- (1) Two
- (2) Three
- (3) Four
- (4) One

21)

The members of phylum porifera are called sponges, having numerous distinguishable character, but which of the following considered as their peculiar character ?

- (1) These are diploblastic animals.
- (2) Most members are marine and some are fresh water.
- (3) Choanocytes line the spongocoel.
- (4) Intracellular digestion.

22) Read the following statements A-D and choose the option which have all the correct ones only. :-

- (A) All vertebrates are chordates but all chordates are not vertebrates.
- (B) In cephalochordata notochord persist throughout the life.
- (C) Protochordates are vertebrates.
- (D) In chordates nerve cord is situated at the ventral surface of body.

- (1) Statement C, D
- (2) Statement A, B
- (3) Statement A,B,D
- (4) Statement B,C and D

23) Members belong to kingdom animalia are characterized by :-

- (1) Heterotrophic, Multicellular
- (2) Eukaryotic organism, cell wall absent
- (3) Directly or indirectly depends upon plants for food.
- (4) All of the above

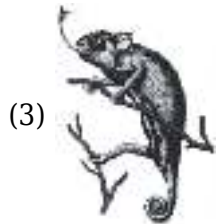
24) Which one of the following pairs of animals are similar to each other pertaining to the feature stated against them?

- (1) *Ascaris* and *Ancylostoma* — Metameric segmentation
- (2) Sea horse and Flying fish — Cold blooded (poikilothermal)
- (3) *Pteropus* and *Ornithorhynchus* — Viviparity
- (4) Garden lizard and Crocodile — Three chambered heart

25) Which one of the following categories of animals, is correctly described with no single exception in it ?

- (1) All sponges are marine and have collar cells.
- (2) All mammals are viviparous and possess diaphragm for breathing.
- (3) All reptiles possess scales, have a three chambered heart and are cold blooded (poikilothermal).
- (4) All bony fishes have gills covered by an operculum on each sides.

26) Which one is poikilothermic and has dryskin ?



27)

Match column-I with column-II and select the correct option :

Column-I		Column-II	
(a)	Phylum	(P)	Agnatha
(b)	Subphylum	(Q)	Cyclostomata
(c)	Division	(R)	Tetrapoda
(d)	Super class	(S)	Chordata
(e)	Class	(T)	Vertebrata

	a	b	c	d	e
(1)	S	T	P	Q	R
(2)	S	T	Q	R	P
(3)	S	T	P	R	Q
(4)	S	R	Q	P	T

(1) 1

(2) 2

(3) 3

(4) 4

28) Birds and mammals have :-

- (1) Three chambered heart with double circulation
- (2) Four chambered heart with double circulation
- (3) Two chambered heart with single circulation
- (4) Four chambered heart with truncus arteriosus

29) Match the column-I and II correctly :-

Column-I		Column-II	
(A)	Mollusca	(i)	Proboscis gland
(B)	Arthropoda	(ii)	Radial symmetry
(C)	Hemichordata	(iii)	Malpighian tubules
(D)	Echinodermata	(iv)	Radula

- (1) A-i, B-iii, C-iv, D-ii
- (2) A-i, B-ii, C-iii, D-iv
- (3) A-i, B-iv, C-ii, D-iii
- (4) A-iv, B-iii, C-i, D-ii

30) Which of the following animals has three chambered heart ?

- (1) *Scoliodon*
- (2) *Hippocampus*
- (3) *Chelone*
- (4) *Pteropus*

31) **Assertion (A):** Binomial nomenclature consists of two words – genus and specific epithet.

Reason (R): This system was given by Ernst Mayr.

- (1) Both A and R are true, and R is the correct explanation of A
- (2) Both A and R are true, but R is not the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

32) Match the organism with the correct scientific name:

Organism	Scientific Name
A. Man	1. <i>Mangifera indica</i>
B. Mango	2. <i>Triticum aestivum</i>
C. Housefly	3. <i>Homo sapiens</i>
D. Wheat	4. <i>Musca domestica</i>

- (1) A-3, B-1, C-4, D-2
- (2) A-1, B-3, C-2, D-4
- (3) A-2, B-4, C-3, D-1
- (4) A-4, B-2, C-1, D-3

33) Scientific name of mango is:

- (1) Mangifera Indica
- (2) *Mangifera indica* Linn
- (3) *mangifera indica*
- (4) Mangifera Indica L.

34) In taxonomy, which term refers to a group at any level?

- (1) Kingdom
- (2) Taxon
- (3) Category
- (4) Species

35) Which one of the following categories has the maximum number of organisms?

- (1) Class
- (2) Order
- (3) Genus
- (4) Kingdom

36) Which of the following is a correct example of genus and specific epithet ?

- (1) *Mangifera - indica*
- (2) *indica - Mangifera*
- (3) *Triticum - aestivum*
- (4) Both (1) and (3)

37) Which of the following organisms belong to order Carnivora?

- (1) Dog, Tiger
- (2) Cow, Goat
- (3) Rabbit, Mouse
- (4) Deer, Buffalo

38) Which of the following refers to the taxonomic category "Phylum"?

- (1) Only used in plants
- (2) Used only for prokaryotes
- (3) Used only for vertebrates
- (4) Used in animals instead of division

39) Which of the following statements are correct regarding binomial nomenclature?

- I.** Genus and species names are always written in Latin.
- II.** The first word in the binomial name starts with a capital letter.
- III.** Scientific names are never underlined even when handwritten.

IV. The naming system was introduced by Carolus Linnaeus.

- (1) Only I, II, and IV
- (2) I, III, and IV
- (3) II and III only
- (4) All are correct

40) Which of the following are true regarding *Mangifera indica*?

- I.** '*Mangifera*' refers to the genus.
- II.** '*indica*' is the specific epithet.
- III.** The name must be underlined if handwritten.
- IV.** It belongs to the family Poaceae.

- (1) I, II, III
- (2) I, II, III, IV
- (3) I and III only
- (4) II, III, IV

41) Lichens are :

- (1) Symbiotic association between algae and fungi
- (2) Pollution indicators
- (3) Not placed in any kingdom by R.H. Whittaker
- (4) All of these are correct

42) Read the following statements :

- (i) Mycorrhiza is the symbiotic association between roots of higher plants and fungus.
- (ii) Lichens are biological indicator for water pollution.
- (iii) Viroids contain only low molecular weight DNA and protein.
- (iv) W.M. Stanley crystallised TMV

Which of the above statements are correct:

- (1) (i), (ii) and (iv)
- (2) (i) and (iv)
- (3) (ii) and (iii)
- (4) (i) and (ii)

43) Match List-I with List-II.

List-I		List-II	
A.	<i>Rhizopus</i>	I.	Mushroom
B.	<i>Ustilago</i>	II.	Smut fungus
C.	<i>Puccinia</i>	III.	Bread mould
D.	<i>Agaricus</i>	IV.	Rust fungus

Choose the correct answer from the options given below :-

- (1) A-III, B-II, C-IV, D-I
 (2) A-I, B-III, C-II, D-IV
 (3) A-III, B-II, C-I, D-IV
 (4) A-IV, B-III, C-II, D-I

44) Match List-I with List-II.

List-I (Organisms)		List-II (Mode of Nutrition)	
A.	Euglenoid	I.	Parasitic
B.	Dinoflagellate	II.	Saprophytic
C.	Slime mould	III.	Photosynthetic
D.	<i>Plasmodium</i>	IV.	Switching between photosynthetic and heterotrophic mode

Choose the correct answer from the options given below :-

- (1) A-III, B-IV, C-II, D-I
 (2) A-IV, B-II, C-I, D-III
 (3) A-IV, B-III, C-II, D-I
 (4) A-IV, B-II, C-III, D-I

45) The following features belongs to class :

- A. Mycelium branched and septate.
 B. Asexual spores conidia produced exogenously.
 C. Sexual spores produced endogenously.
 D. Many members of this class are edible.

- (1) Ascomycetes
 (2) Phycomycetes
 (3) Basidiomycetes
 (4) Deuteromycetes

46) **Assertion** : Euglenoids have flexible body.

Reason : Euglenoids are covered by protein rich layer called pellicle.

- (1) Both assertion and reason are true and reason is a correct explanation of assertion.
 (2) Both assertion and reason are true and reason is not a correct explanation of assertion.
 (3) Assertion is true but reason is false.
 (4) Both assertion and reason are false.

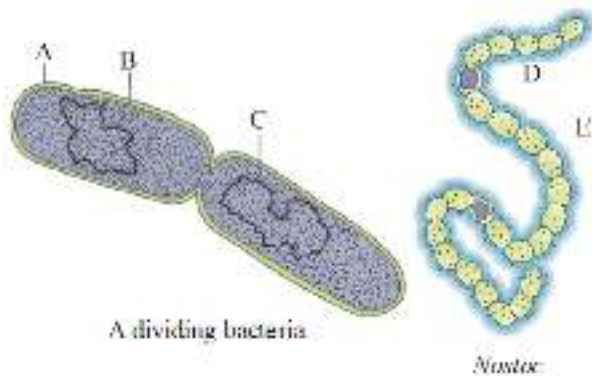
47) Match the following :

A	Phycomycetes	(i)	Club fungi
B	Ascomycetes	(ii)	Fungi imperfecti

C	Basidiomycetes	(iii)	Sac fungi
D	Deuteromycetes	(iv)	Algal fungi

- (1) A - (i), B - (ii), C - (iii), D - (iv)
 (2) A - (i), B - (iii), C - (ii), D - (iv)
 (3) A - (iv), B - (iii), C - (i), D - (ii)
 (4) A - (iv), B - (i), C - (iii), D - (ii)

48) Identify the A, B, C, D and E in the following diagram :



- (1) A - Cell membrane, B - Cell wall, C - DNA, D - Heterocyst, E - Mucilagenous sheath
 (2) A - Mucilagenous sheath, B - Cell membrane, C - DNA, D - Heterocyst, E - Cell wall
 (3) A - Cell wall, B - Cell membrane, C - DNA, D - Heterocyst, E - Mucilagenous sheath
 (4) A - Cell wall, B - Cell membrane, C - Heterocyst, D - DNA, E - Mucilagenous sheath

49) In earlier classification (like two kingdoms), the following are included in plant

- (I) Bacteria, blue green algae, fungus
 (II) Mosses and fern
 (III) Gymnosperms and angiosperm

- (1) (I) only
 (2) (II) and (III) only
 (3) (III) only
 (4) All of these

50) According to R.H. Whittaker, *Chlamydomonas* & *Chlorella* are placed in Kingdom :-

- (1) Plantae
 (2) Animalia
 (3) Protista
 (4) Monera

51) A unicellular eukaryotic wall less microbe was collected by a student from rain water. His teacher grouped the organism in ____ kingdom according to Linnaeus system and ____ kingdom according to Whittaker's system.

- (1) Animalia, monera
 (2) Monera, plantae

- (3) Plants, protista
- (4) Animalia, protista

52) Cellulosic cell wall is absent in :-

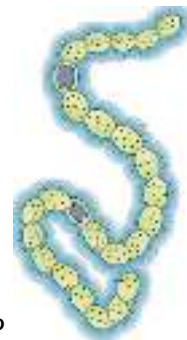
- (1) Most of the fungi, Euglenoids & Bacteria
- (2) All fungi, Dinoflagellates and Bacteria
- (3) Dinoflagellates, Diatoms and slime moulds
- (4) Bacteria, Diatoms and Desmids

53) Which of the following is wrong statement :

- (1) Bacterial structure is very simple but they are very complex in behaviour.
- (2) Chemosynthetic nature for mode of nutrition is unique in bacteria.
- (3) Archaeobacteria differ from other bacteria in absence of cell membrane.
- (4) Most of cyanobacteria are photosynthetic autotrophs.

54) Beside paddy fields cyanobacteria (BGA) are also found inside vegetative part of :-

- (1) *Psilotum*
- (2) *Pinus*
- (3) *Cycas*
- (4) *Equisetum*



55) How many characters in the list given below are correct for this diagram ?

(i)

Non-oxygenic photosynthesis

- (ii) N_2 -fixation
- (iii) Decomposer
- (iv) Stored food similar to plants
- (v) Production of antibiotics
- (vi) Peptidoglycan cell wall

Options :

- (1) i, ii, iv
- (2) ii, vi
- (3) ii, iv, v, vi
- (4) i, ii, iii, v

56) How many statements is/are incorrect about mycoplasma :-

- (A) Cell wall absent

- (B) Called joker of plant kingdom
- (C) Resistant to tetracycline & chloramphenicol
- (D) Osmotrophic mode of nutrition
- (E) Cell membrane is trilayered
- (F) Facultative parasite

- (1) A, B, F
- (2) A, C, D, E
- (3) Only C
- (4) Only F

57) Which of the following characters are true for slime moulds ?

- (1) Saprophytic, plasmodium, spores dispersed by water
- (2) Parasitic, plasmodium, spores dispersed by water
- (3) Saprophytic, plasmodium, spores dispersed by air currents
- (4) Parasitic, plasmodium, spores dispersed by insects.

58) Match Column-I with Column-II and select the correct option

Column-I		Column-II	
(A)	Production of methane	(i)	Methanogen bacteria
(B)	Most extensive metabolic diversity	(ii)	Halophiles
(C)	Hot spring	(iii)	Monera
(D)	Extreme salty area	(iv)	Thermoacidophiles

- (1) A-i, B-iii, C-iv, D-ii
- (2) A-ii, B-i, C-iii, D-iv
- (3) A-iii, B-ii, C-i, D-iv
- (4) A-iv, B-iii, C-ii, D-i

59) Match the column-I with column-II and choose the right option :-

Column-I		Column-II	
A.	Ascomycetes	(i)	<i>Trichoderma</i>
B.	Deuteromycetes	(ii)	<i>Albugo</i>
C.	Phycomycetes	(iii)	<i>Claviceps</i>
D.	Basidiomycetes	(iv)	<i>Agaricus</i>

- (1) A-i, B-ii, C-iii, D-iv
- (2) A-ii, B-i, C-iv, D-iii

(3) A-iii, B-i, C-ii, D-iv

(4) A-iv, B-iii, C-i, D-ii

60) Which fungi cause rust and smut diseases respectively ?

(1) *Ustilago* and *Agaricus*

(2) *Puccinia* and *Ustilago*

(3) *Puccinia* and *Agaricus*

(4) *Ustilago* and *Puccinia*

61) Sexual reproduction in fungi may occur by means of :-

(1) Sporangiospore, Oospore and Ascospore

(2) Zoospore, Oospore and Ascospore

(3) Sporangiospore, Ascospore and basidiospore

(4) Oospore, Ascospore and basidiospore

62) "Late blight of potato" Disease is caused by:-

(1) A member of Phycomycetes

(2) A member of Ascomycetes

(3) A member of basidiomycetes

(4) A member of deuteromycetes

63) Who demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants ?

(1) Ivanowsky

(2) Beijerinck

(3) Stanley

(4) Pasteur

64) In humans Cr-Jacob disease (CJD) is caused by:-

(1) Prion

(2) Viroid

(3) Virus

(4) PPLO

65)

Prions an agent consist of :-

(1) Normally folded protein

(2) RNA + Protein

(3) DNA + Protein

(4) Abnormally folded protein

66) *Ulothrix*, *Spirogyra* and *Chara* are member of :-

- (1) Pheophyceae
- (2) Chlorophyceae
- (3) Rhodophyceae
- (4) Cyanophyceae

67) Which of the following is motile and flagellate ?

- (1) Zoospore
- (2) Akinete
- (3) Aplanospore
- (4) Hypnospore

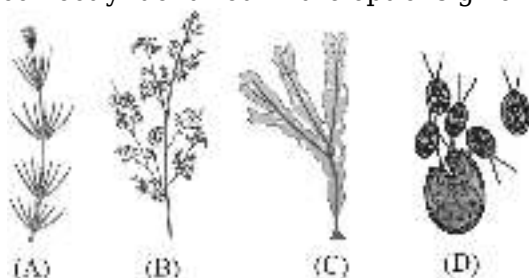
68) Which of the following is not the basis of classification in algae ?

- (1) Pigment
- (2) Cell wall composition
- (3) Stored food
- (4) Place of occurrence

69) All given algal members possess unicelled sex organ except :-

- (1) *Chara*
- (2) *Ulothrix*
- (3) *Spirogyra*
- (4) *Chlamydomonas*

70) The four diagrams are (A,B,C and D) given below represent four different types of organism. Which one of these is correctly identified in the options given along with its correct stored food and



shape of chloroplast ? (A)

(B)

(C)

(D)

		Organism	Stored food	Shape of chloroplast
(1)	D	<i>Chlamydomonas</i>	Starch	Cup shaped
(2)	C	<i>Fucus</i>	Algin	Cup shaped
(3)	B	<i>Porphyra</i>	Floridean starch	Plate like

(4)	A	<i>Chara</i>	Starch	Spiral shape
-----	---	--------------	--------	--------------

- (1) 1
(2) 2
(3) 3
(4) 4

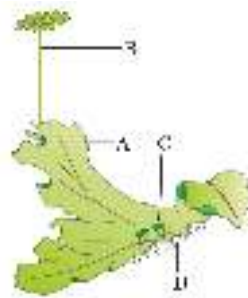
71) "Gemmae" are multicellular green structures for vegetative propagation. These are found in gemma cups in ?

- (1) *Riccia* thallus
(2) *Marchantia* thallus
(3) *Funaria* protonema
(4) Fern prothallus

72) Archegoniophore is present in :-

- (1) *Marchantia*
(2) *Chara*
(3) *Adiantum*
(4) *Funaria*

73) Examine the figure given below and select the right option in which all the four labelled parts (A,



B, C, D) are correctly identified :-

	A	B	C	D
(1)	Female Thallus	Archegoniophore	Gemma cup	Rhizoids
(2)	Male Thallus	Antheridiophore	Gemma cup	Rhizoids
(3)	Female Thallus	Antheridiophore	Rhizoids	Gemma cup
(4)	Male Thallus	Archegoniophore	Rhizoids	Gemma cup

- (1) 1
(2) 2
(3) 3
(4) 4

74) Spores do not form protonema but directly grow into flat branching thallus in :-

- (1) Liverworts

- (2) Mosses
- (3) Both (1) & (2)
- (4) None

75) Protonema is generally produced in the life cycle of :

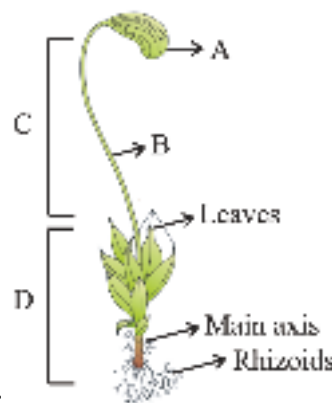
- (1) Liver wort
- (2) Horn wort
- (3) Mosses
- (4) Both (1) and (2)

76) In moss sporophyte, which of the following is absent ?

- (1) Foot
- (2) Seta
- (3) Capsule
- (4) Elaters

77) When *Funaria* spores germinate, they form :-

- (1) Leafy gametophyte directly
- (2) Capsule directly
- (3) Protonema, which bears archegonia and antheridia
- (4) First protonema, then forms a bud and then a leafy gametophyte



78) In given figure A B, C and D represent :

- (1) A-Sporangium, B-Capsule, C-Sporophyte, D-Gametophyte
- (2) A-Capsule, B-Seta, C-Sporophyte, D-Gametophyte
- (3) A-Sporangium, B-Seta, C-Gametophyte, D-Sporophyte
- (4) A-Sporangium, B-Capsule, C-Gametophyte, D-Sporophyte

79) Which one of the following is homosporous ?

- (1) *Azolla*
- (2) *Equisetum*
- (3) *Cedrus*
- (4) *Salvinia*

80) Prothallus is present in :-

- (1) *Polytrichum*
- (2) *Funaria*
- (3) *Marchantia*
- (4) *Pteris*

81) Fertilization takes place by spiral and multiflagellated antherozoids in :-

- (1) Ferns
- (2) Mosses
- (3) Liverwort
- (4) Gnetales

82) Which of the following is responsible for peat formation ?

- (1) *Marchantia*
- (2) *Riccia*
- (3) *Funaria*
- (4) *Sphagnum*

83) In fern, prothallus develops :-

- (1) By fertilisation
- (2) From zygote
- (3) By secondary protonema
- (4) From spore

84) Which statement is wrong for *Cycas* ?

- (1) Xylem have vessels
- (2) Male cones are well developed
- (3) It has coralloid roots
- (4) Circinate ptyaxis/Circinate vernation

85) *Cycas* have two cotyledons but not included in angiosperms because of :-

- (1) Naked ovules
- (2) Seems like monocot
- (3) Circinate ptyxis
- (4) Compound leaves

86) Read the following statements & select the correct option :

- (A) Gymnosperms do not show xerophytic characters.
- (B) In *Cycas* coralloid roots are associated with mycorrhiza
- (C) In conifers needle like leaves reduce the surface area.
- (D) All gymnosperms are heterosporous

How many above statements are correct & incorrect?

- (1) 2 - correct, 2 - incorrect
- (2) 3 - correct, 1 - incorrect
- (3) 1 - correct, 3 - incorrect
- (4) 0 - correct, 4 - incorrect

87) **Assertion** : In gymnosperms seeds are naked.

Reason : In gymnosperms seeds are not protected with fruit covering.

- (1) Both assertion and reason are correct and reason is the correct explanation of assertion.
- (2) Both assertion and reason are correct but reason is not the correct explanation of assertion.
- (3) Assertion is correct but reason is incorrect.
- (4) Both assertion and reason are incorrect.

88) Choose the incorrect statement from the following :

- (1) Almost all Gymnosperms are Xerophytes
- (2) All Gymnosperms show Heterosporous nature
- (3) All Gymnosperms show seed habit
- (4) All Gymnosperms are Monoecious

89) The functional megaspore of gymnosperm germinate :-

- (1) Inside the megaspore mother cell
- (2) Inside the megasporangium
- (3) Inside the soil
- (4) Into endosperm after fertilization

90) Read the following statements :-

- (A) *Selaginella* and *Salvinia* are the heterosporous pteridophytes because they produce two different kinds of spores.
- (B) The gametophytes of homosporous pteridophytes bear male and female sex organs called antheridia and archegonia respectively.
- (C) The female gametophytes in some pteridophytes are retained on the parent sporophytes, this event is a precursor to seed habit.
- (D) Pteridophytes are first terrestrial plants and they possess seeds.

How many of the above statement(s) is/are not correct ?

- (1) Four
- (2) Three

(3) Two

(4) One

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	4	3	4	2	2	3	4	2	3	2	2	1	3	1	3	1	4	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	1	2	1	1	4	4	1	1	1	2	2	1	2	1	4	3	2	2
Q.	41	42	43	44	45															
A.	3	1	2	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	4	4	2	1	1	4	2	3	1	1	1	2	2	2	4	3	3	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	2	4	2	2	2	1	4	2	2	3	1	3	3	2	2	2	3	1
Q.	86	87	88	89	90															
A.	1	1	3	3	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	3	2	2	3	3	1	1	4	4	4	3	2	2	3	4	2	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	4	2	4	3	3	2	4	3	3	1	2	2	4	4	1	4	1	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	1	3	1	1	3	3	4	3	4	1	3	3	2	3	3	1	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	1	2	1	4	2	1	4	1	1	2	1	2	1	3	4	4	2	2	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	4	4	1	1	1	1	4	2	4										

SOLUTIONS

PHYSICS

1) **Question Explanation :** Two parallel vectors is given, calculate value of $m \times n$.

Concept: Parallel vectors.

Solution:

$$\vec{a} = 2\hat{i} + 6\hat{j} + m\hat{k}$$

$$\vec{b} = n\hat{i} + 12\hat{j} + 4\hat{k}$$

For parallel vectors

$$\frac{2}{n} = \frac{6}{12} = \frac{m}{4}$$

$$n = 4 \quad m = 2, \Rightarrow mn = 8$$

Final answer: option (3)

2) **Explain Question:**

we have to calculate change in velocity

Concept:

subtraction of vectors

Solution:

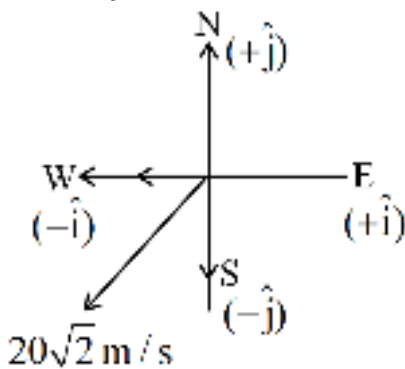
$$V_i = 20\hat{j}$$

$$V_f = 20(-\hat{i})$$

$$\Delta V = V_f - V_i$$

$$= -20\hat{i} - 20\hat{j} = 20\sqrt{2}$$

$$|\Delta V| = \sqrt{(20)^2 + (20)^2} = 20\sqrt{2} \text{ m/s}$$



Final answer: (4)

3)

Explain the Question : Find $\int 2 \sin(2x + 3) dx$ using standard integral formulas.

Concept : This question is based on use formula : $\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C$.

Solution : $\int 2 \sin(2x + 3) dx = 2 \int \sin(2x + 3) dx = 2 \left(-\frac{1}{2} \cos(2x + 3) \right) + C = -\cos(2x + 3) + C$.

Final answer : 3

4) **Explaining :** Choose correct option

Concept : Trigonometry

Solution : $\cos(\alpha + \beta) - \cos(\alpha - \beta)$
 $= \cos\alpha \cos\beta - \sin\alpha \sin\beta - [\cos\alpha \cos\beta + \sin\alpha \sin\beta]$
 $= -2 \sin\alpha \sin\beta$

Final Answer : (4)

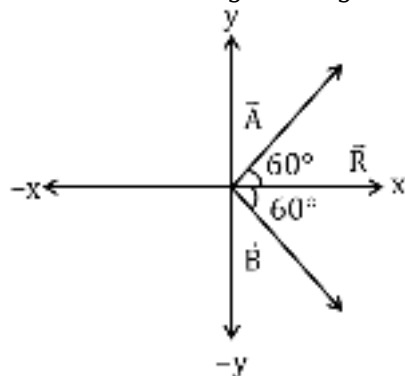
5) **Explaining :** $|\vec{A}| = 2$ cm. making 60° from +x-axis in anticlockwise direction

$|\vec{B}| = 2$ cm, making 60° from +x-axis in clockwise direction

ASK : - Magnitude and direction of $\vec{A} + \vec{B}$

Concept : This question is based on if $|\vec{A}|$ and $|\vec{B}|$ and θ is angle between $\vec{A}$ and $\vec{B}$ then
 $|\vec{R}| = |\vec{A} + \vec{B}| = 2A \cos\left(\frac{\theta}{2}\right)$

where direction of $\vec{R}$ is along the angle bisector of $\vec{A}$ and $\vec{B}$



Solution :

Both vectors have equal magnitude, $\theta = 120^\circ$

So, the resultant will be of same magnitude in positive x direction.

Final Answer : (2) 2cm along +x-axis

6) **Explaining :**

Given : - $v = (t + 2)(t + 3)$

ASK : - acceleration at $t = 1$ s

Concept : This question is based on $a = \frac{dv}{dt}$

Solution : $v = (t + 2)(t + 3) = t^2 + 5t + 6$

$a = \frac{dv}{dt} = 2t + 5$

at $t = 1$ s :-

$$a = 2(1) + 5 = 7 \text{ m/s}^2$$

Final Answer : (2) 7 m/s^2

7) Question Explanation:

We are asked to find the roots of the quadratic equation $10x^2 - 27x + 5 = 0$.

Concept: This question is based on Quadratic equation.

Formula :

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solution:

Here, $a = 10$, $b = -27$, and $c = 5$. Applying the quadratic formula:

Calculation :

$$\text{Discriminant} = b^2 - 4ac = (-27)^2 - 4(10)(5) = 529$$

$$x = \frac{27 \pm \sqrt{529}}{20} = \frac{27 \pm 23}{20} \quad x_1 = \frac{5}{2} \quad x_2 = \frac{1}{5}$$

Final Answer: (3)

8) Explaining : To find the combination of force whose resultant is non-zero.

Concept : This question is based on

Three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ where $\vec{C}$ has greatest magnitude among them, can give zero resultant if

$$||\vec{A}| - |\vec{B}|| \leq |\vec{C}| \leq |\vec{A}| + |\vec{B}|$$

Solution : $A - B \leq C \leq A + B$

For $A = 10$, $B = 10$, $C = 10$

$$10 - 10 \leq 10 \leq 10 + 10$$

(10, 10, 10) can produce zero resultant same is possible for (10, 10, 20) and (10, 20, 20)

But 10, 20, 40 when $C = 40$, $A = 10$, $B = 20$

$$A + B = 30 < C$$

Therefore 10, 20, 40 can not produce zero resultant

Final Answer : (4) 10, 20, 40

9) Explaining : If $y = 2x^2 - 2x$ then find out minimum value of y :

Concept : Application of differentiation

Solution :

$$y = 2x^2 - 2x$$

$$\Rightarrow \frac{dy}{dx} = 4x - 2 = 0$$

$$x = 1/2$$

$$\frac{d^2y}{dx^2} = 4 > 0$$

so min is at $x = 1/2$

$$y_{\min} = (2) \left(\frac{1}{2}\right)^2 - 2 \left(\frac{1}{2}\right)$$

$$= -1/2$$

Final Answer : (2)

10) **Explaining** : To check if Assertion and Reason are correct and if reason explains Assertion.

Concept : This question is based on

If $\vec{A}$ and $\vec{B}$ are two vector then

Cross product : $\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta \hat{n}$ is a vector quantity

Dot product : $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$ is a scalar quantity

Solution : $\vec{A} \times \vec{B}$ is vector and $\vec{A} \cdot \vec{B}$ is scalar cannot compare. So, Assertion is wrong. Answer is (3).

Final Answer : (3) Assertion is wrong, reason is correct.

11) **Explanation**: Find the components of the vector along the new x' and y' axes

Concept: Resolution of vector

Solution:

New angle $\theta' = 37^\circ + 16^\circ = 53^\circ$

x'-component $5 \cos 53^\circ = 5 \times \frac{3}{5} = 3\text{m}$

y'-component $5 \sin 53^\circ = 5 \times \frac{4}{5} = 4\text{m}$

Final Answer: (2) (3,4)

12) **Explanation**: Find the projection of $\hat{i} + 2\hat{j}$ on $2\hat{i} + 3\hat{j}$ in vector form

Concept: Application of dot product

Solution:

Let $\vec{A} = \hat{i} + 2\hat{j}$, $\vec{B} = 2\hat{i} + 3\hat{j}$

Projection of $\vec{A}$ on $\vec{B}$ in vector form

$$\left(\frac{\vec{A} \cdot \vec{B}}{B^2} \right) \vec{B} = \frac{(\hat{i} + 2\hat{j}) \cdot (2\hat{i} + 3\hat{j})}{\{\sqrt{2^2 + 3^2}\}^2} (2\hat{i} + 3\hat{j})$$

$$= \frac{8}{13} (2\hat{i} + 3\hat{j})$$

Final Answer: (2)

13) **Explanation**: Question asking for correct statements regarding the vector properties

Concept: Properties of vector addition, product

Solution:

- Vector addition is commutative
- $\vec{A} - \vec{B} = -(\vec{B} - \vec{A})$
- Dot product zero $\Rightarrow$ vectors can be perpendicular (not necessarily zero)
- Cross product is anti-commutative

Final Answer: (1)

14) **Explaining** : Find diameter of cylinder.

Concept : This question is based on vernier callipers.

Solution :

10 VSD = 9 MSD

$$\text{L.C.} = 0.1 \text{ mm}$$

$$= 0.01 \text{ cm}$$

$$+ZE = 5 \times 0.1 \text{ mm} = 0.5 \text{ mm} = 0.05 \text{ cm}$$

$$\text{Reading} = 3.1 \text{ cm} + 4 (0.01 \text{ cm}) - 0.05 \text{ cm}$$

$$\text{Reading} = 3.09 \text{ cm}$$

Final Answer : (3)

15) **Explaining** : Find relation b/w T, P, d & E.

Concept : Relation b/w diff. physical quantities using dimensional analysis.

Solution :

$$T = KP^a d^b E^c$$

$$[T] = [ML^{-1}T^{-2}]^a [ML^{-3}]^b [ML^2T^{-2}]^c$$

$$a + b + c = 0$$

$$-a - 3b + 2c = 0$$

$$-2a - 2c = 1$$

$$a + c = -\frac{1}{2}$$

$$a = -\frac{5}{6}, b = \frac{1}{2}, c = \frac{1}{3}$$

$$\text{So, } T \propto P^{-5/6} d^{1/2} E^{1/3}$$

Final Answer : (1)

16)

Explain the Question : A screw gauge

$$s = 0.5 \text{ mm}$$

$$N_{\text{rotation}} = 1$$

$$N_{\text{circular}} = 50$$

$$\text{MSR} = 2.5 \text{ mm}$$

$$\text{CSR} = 45 \text{ CSD}$$

$$\text{ZE} = -0.03 \text{ mm}$$

$$\text{NR} = ?$$

Concept : This question is based on screw gauge.

Solution :

$$\text{Pitch} = \frac{S}{N_{\text{rotation}}} = \frac{0.5 \text{ mm}}{1} = 0.5 \text{ mm}$$

$$\text{Least count} = \text{LC} = \frac{P}{N_{\text{circular}}} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$$

$$\text{Net reading} =$$

$$\text{NR} = \text{MSR} + (\text{CSR} \times \text{LC}) - \text{ZE}$$

$$= 2.5 \text{ mm} + (45 \times 0.01 \text{ mm}) - (-0.03 \text{ mm})$$

$$= 2.5 \text{ mm} + 0.45 \text{ mm} + 0.03 \text{ mm}$$

$$= 2.98 \text{ mm}$$

Final answer : 3

17) **Explaining** : Find dimensionally correct formula for terminal velocity.

Concept : This question is based on dimensional correctness.

Solution :

$$\text{L.H.S. } [V] = [M^0 L^1 T^{-1}]$$

RHS for option-(1)

$$\frac{[L^2] [L^1 T^{-2}] [M^1 L^{-3}]}{[M^1 L^{-1} T^{-1}]} = [L^1 T^{-1}]$$

Final Answer : (1)

18) **Explaining** : Find correct option for given Assertion & Reason.

Concept : This question is based on units.

Solution :

Light year :- distance travelled by light in one year.

Parallactic second :- 3.08×10^{16} m it is a length.

Final Answer : (4)

19) **Explaining** : Find percentage error in density.

Concept : This question is based on propagation in error.

Solution : Density (ρ) = $\frac{\text{mass}(m)}{\text{volume}(V)}$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$$

$$\frac{\Delta \rho}{\rho} = \left(\frac{0.05}{5}\right) 100\% + \left(\frac{0.05}{1}\right) 100\%$$

$$= 1\% + 5\% = 6\%$$

Final Answer : (3)

20) **Explaining** : Find incorrect option.

Concept : This question is based on principle of homogeneity.

Solution :

$[A] = [B] = [A + B]$ = Dimension of length

$[A \cdot B]$ = Dimension of area, $\left[\frac{AB}{C}\right] \rightarrow$ Dimension of length.

$\frac{[A]}{[B + C]}$ = Dimension-less

Final Answer : (1)

21) **Explaining** : Find values of x and y.

Concept : This question is based on dimensional analysis.

Solution :

$$f = cm^x k^y$$

$$\text{As } [f] = [M^0 L^0 T^{-1}], [M] = [M^1 L^0 T^0]$$

$$\text{and } [k] = [M^1 L^0 T^{-2}]$$

$$\text{so } [M^0 L^0 T^{-1}] = [M^1 L^0 T^0]^x [M^1 L^0 T^{-2}]^y$$

$$\Rightarrow [M^0 L^0 T^{-1}] = [M^{x+y} L^0 T^{-2y}]$$

By comparing we get $x + y = 0$ and $-2y = -1$

$$\text{so } y = \frac{1}{2} \text{ \& } x = -\frac{1}{2}$$

Final Answer : (4)

22) **Explaining** : Find value of power in given system of unit.

Concept : Change in system of unit using dimensions.

Solution :

$$n_2 = 1 \left[\frac{1 \text{ kg}}{20 \text{ kg}} \right]^1 \left[\frac{1 \text{ m}}{10 \text{ m}} \right]^2 \left[\frac{1 \text{ sec}}{5 \text{ sec}} \right]^{-3} = \frac{1}{20} \times \frac{1}{100} \times 125 = \frac{1}{16}$$

$\frac{1}{16}$ unit = 1 watt

1 unit = 16 watts.

Final Answer : (1)

23) **Explaining** : Match the column of physical quantity with their dimensions.

Concept : This question is based on dimensions of physical quantity.

Solution :

$$[\text{Torque}] = [M^1 L^2 T^{-2}]$$

$$[\text{Angular momentum}] = [M^1 L^2 T^{-1}]$$

$$[\text{latent heat}] = [M^0 L^2 T^{-2}]$$

$$[\text{surface tension}] = [M^1 L^0 T^{-2}]$$

$$[\text{light year}] = [M^0 L^1 T^0]$$

Final Answer : (1)

24) **Explaining** : Find percentage error in Z.

Concept : This question is based on propagation in error.

Solution :

$$z = a^2 x^3 y^{1/2}$$

$$\frac{\Delta z}{z} = \frac{2\Delta a}{a} + \frac{3\Delta x}{x} + \frac{1}{2} \frac{\Delta y}{y}$$

a is constant

$$\frac{\Delta z}{z} \times 100 = 3(4\%) + \frac{1}{2}(12\%) = 18\%$$

Final Answer : (2)

25) **Explaining** : Find correct option for given Assertion & Reason.

Concept : This question is based on significant figure.

Solution :

Rules in Reason are correct

$$0.005 = 1 \text{ S.F.}$$

$$0.500 = 3 \text{ S.F.}$$

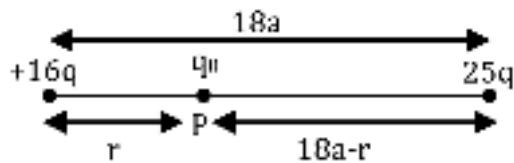
Final Answer : (1)

26) **Explaining** : To locate the location of charge q_0

So that it experiences zero net force.

Concept : This question is based on equilibrium of charges.

Solution :



For q_0 to be in equilibrium

$$\frac{k(16q)(q_0)}{r^2} = \frac{k(25q)(q_0)}{(18a-r)^2}$$

taking square root both sides

$$\frac{4}{r} = \frac{5}{18a-r}$$

$$72a - 4r = 5r$$

$$8a = r$$

$$x = 10a$$

Final Answer : (1)

27) **Explaining :** To determine the correct equilibrium configuration.

Concept : This question is based on suspended charge system

$$\text{Here } \tan \theta = \frac{F}{mg}$$

Solution :

When the charge on one pith ball is doubled, the repulsive force between the balls increases, causing them to move further apart until a new equilibrium is reached, and the angle of the thread with the vertical will increase.

Explanation:

1. Equal Charges:

When both pith balls have an equal positive charge (Q), they repel each other. The repulsive force is balanced by the tension in the suspension threads and the force of gravity on each ball.

2. Increased Charge:

If the charge on one ball is doubled to $2Q$, the repulsive force between the balls increases because the force is directly proportional to the product of the charges.

3. Increased Separation:

The stronger repulsive force pushes the balls further apart to achieve a new equilibrium where the forces are again balanced.

4. Angle Increase:

Since the balls are now further apart, the angle each thread makes with the vertical will increase to maintain the balance of forces.

Final Answer : (4)

28) **Explaining :**

Concept : This question is based on

Solution :

Coulomb's Law (Vector Form):

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^3} \cdot \vec{r}$$

Where:

- $\vec{r} = \vec{r}_2 - \vec{r}_1$ is the displacement vector from q_2 to q_1
- $r = |\vec{r}|$ is the distance between the charges

- $\vec{F}_{21}$ is the force **on** q_2 **due to** q_1

Step-by-Step:

1. Position Vectors:

- $\vec{r}_1 = 2\hat{i} - \hat{j} + 3\hat{k}$
- $\vec{r}_2 = 0\hat{i} + 0\hat{j} + 0\hat{k}$
- $\vec{r} = \vec{r}_2 - \vec{r}_1 = -2\hat{i} + \hat{j} - 3\hat{k}$

2. Distance r:

$$r = \sqrt{(-2)^2 + 1^2 + (-3)^2} = \sqrt{4 + 1 + 9} = \sqrt{14}$$

3. Unit vector:

$$\hat{r} = \frac{\vec{r}}{r} = \frac{-2\hat{i} + \hat{j} - 3\hat{k}}{\sqrt{14}}$$

4. Coulomb Force Vector:

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} \cdot \hat{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{14} \cdot \frac{1}{\sqrt{14}} (-2\hat{i} + \hat{j} - 3\hat{k})$$

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{14\sqrt{14}} (-2\hat{i} + \hat{j} - 3\hat{k})$$

Final Answer : (4)

29) **Explaining** : We are given:

- Two **negative charges**, each of magnitude $-q$, placed at a distance $2r$ apart.
- A **positive charge** $+q$ is placed at the **midpoint** between them (i.e., at distance r from each).
- The **initial potential energy** of the system is U_1 .

Then, the **positive charge** and one of the **negative charges** are **interchanged**, forming a new configuration.

The **new potential energy** is U_2 .

We are to find the **ratio** $\frac{U_1}{U_2}$.

Concept : This question is based on

The **potential energy** of a system of two point charges is:

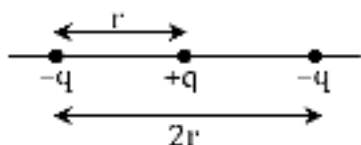
$$U = \frac{kq_1 q_2}{r}$$

We will calculate:

- U_1 — for the original configuration
- U_2 — for the interchanged configuration

and then compute the ratio $\frac{U_1}{U_2}$

Solution : Initial Configuration (for U_1):



Compute all pairwise potential energies:

$$U_1 = U_{(-q, +q)} + U_{(+q, -q)} + U_{(-q, -q)}$$

$$\frac{k(-q)(+q)}{r} + \frac{k(+q)(-q)}{r} + \frac{k(-q)(-q)}{2r}$$

$$= -\frac{kq^2}{r} - \frac{kq^2}{r} + \frac{kq^2}{2r} = -\frac{2kq^2}{r} + \frac{kq^2}{2r} = -\frac{3kq^2}{2r}$$

So,

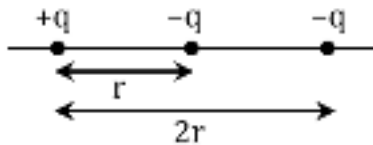
$$U_1 = -\frac{3kq^2}{2r}$$

$$\frac{U_1}{U_2} = -\frac{\frac{3kq^2}{2r}}{\frac{-kq^2}{2r}} = 3$$

So,

After Interchange (for U_2):

Let's say we swap the **left** negative charge and the **middle** positive charge.



Now compute all interactions:

$$U_2 = \frac{k(+q)(-q)}{r} + \frac{k(-q)(-q)}{r} + \frac{k(+q)(-q)}{2r}$$

$$= -\frac{kq^2}{r} + \frac{kq^2}{r} - \frac{kq^2}{2r} = 0 - \frac{kq^2}{2r}$$

So,

$$U_2 = -\frac{kq^2}{2r}$$

$$\frac{U_1}{U_2} = \frac{-\frac{3kq^2}{2r}}{-\frac{kq^2}{2r}} = 3$$

So,

Final Answer : (1)

30) **Explaining :** For net field at O to be along y direction calculate $\frac{\lambda_1}{\lambda_2}$.

Concept : This question is based on field due to semi-infinite wire

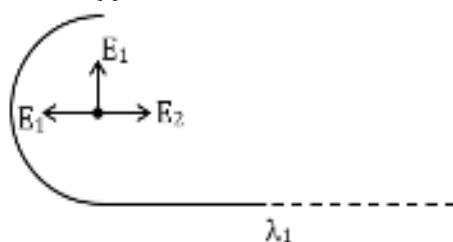
$$E = \frac{\sqrt{2}k\lambda}{r}$$

field due to semi-circular arc $E = \frac{2k\lambda}{r}$

Solution :

$$E_1 = \frac{k\lambda_1}{R}$$

$$E_2 = \frac{2k\lambda_2}{R}$$



$$k\frac{\lambda_1}{R} = 2\frac{k\lambda_2}{R}$$

$$\lambda_1 = 2\lambda_2$$

Final Answer : (1)

31) **Question Explanation:**

Sketch the electric field lines for three equal positive charges placed at the vertices of an equilateral triangle.

Concept:

Properties of electric field lines.

Solution:

- Each charge emits field lines radially outward.
- The line bend away from the triangle center due to mutual repulsion.
- The correct sketch shows symmetrical outward line, no convergence or loops.



Final Answer: Option (1)

32) **Explaining :** To find the no. of electrons in excess or deficit on each sphere.

Concept : This question is based on electric charge and quantisation in charge

Solution :

$$\begin{aligned} q_1 &= q_2 = \left(\frac{-9 + 5}{2} \right) \mu\text{C} \\ &= -2 \mu\text{C} \\ &= \frac{q}{e} = \frac{2 \times 10^{-6}}{1.6 \times 10^{-19}} \\ &= 1.25 \times 10^{13} \text{ electron} \end{aligned}$$

Final Answer : (2)

33) **Explaining :** We are given:

- A **solid sphere** of radius R , with total charge Q , uniformly distributed throughout its **volume**.

R

- A **point charge** $+q$ (mass m) is placed at distance $\frac{R}{2}$ from the center of the sphere.
- It is released from rest and moves under **electrostatic force only**.
- We're asked to find the **speed** of the point charge when it reaches a point **at a distance $2R$ from the center**

(since it's R from the **surface**).

Concept : This question is based on

We'll apply **conservation of mechanical energy**:

$$U_{\text{initial}} + K_{\text{initial}} = U_{\text{final}} + K_{\text{final}}$$

- Initial kinetic energy = 0 (released from rest)
- So:

$$U_i = U_f + \frac{1}{2}mv^2 \Rightarrow \frac{1}{2}mv^2 = U_i - U_f$$

The **potential energy** is due to the electric potential caused by the **charged sphere**. Since

the charge distribution is **uniform**, the electric potential depends on whether the point is **inside** or **outside** the sphere:

- **Inside** the uniformly charged solid sphere (at $r < R$):

$$V(r) = \frac{kQ}{2R^3}(3R^2 - r^2)$$

- **Outside** the sphere (at $r \geq R$):

$$V(r) = \frac{kQ}{r}$$

And potential energy of charge

$$U = qV$$

Solution : Let's find potential energy at both points:

Initial position: $r_i = \frac{R}{2}$ (inside the sphere)

$$V_i = \frac{kQ}{2R^3} \left(3R^2 - \left(\frac{R}{2} \right)^2 \right) = \frac{kQ}{2R^3} \left(3R^2 - \frac{R^2}{4} \right) = \frac{kQ}{2R^3} \left(\frac{11R^2}{4} \right) = \frac{11kQ}{8R}$$

So,

$$U_i - qV_i = \frac{11kQq}{8R}$$

Final position: $r_f = 2R$ (outside the sphere)

$$V_f = \frac{kQ}{2R} \rightarrow U_f = qV_f = \frac{kQq}{2R}$$

Use energy conservation:

$$\frac{1}{2}mv^2 = U_i - U_f = \frac{11kQq}{8R} - \frac{kQq}{2R} = \frac{11kQq - 4kQq}{8R} = \frac{7kQq}{8R}$$

$$\Rightarrow v = \sqrt{\frac{2}{m} \left(\frac{7kQq}{8R} \right)} = \sqrt{\frac{7kQq}{4Rm}}$$

Final Answer : (2)

34) **Explaining :** You're given an electric potential function:

$$V(x, y, z) = x^2y + yz$$

You are asked to find the **magnitude of the electric field** at the point (1,3,1).

Concept : This question is based on

The electric field $\vec{E}$ is the **negative gradient** of the potential function:

$$\vec{E} = - \left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right) = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

We'll:

- Compute the partial derivatives,
- Evaluate them at the given point,
- Find the magnitude of the vector.

Solution : Given:

$$V(x, y, z) = x^2y + yz$$

Compute partial derivatives:

$$\frac{\partial V}{\partial x} = 2xy$$

$$\frac{\partial V}{\partial y} = x^2 + z$$

$$\bullet \frac{\partial V}{\partial z} = y$$

At point (1, 3, 1):

$$\bullet \frac{\partial V}{\partial x} = 2(1)(3) = 6 \Rightarrow E_x = -6$$

$$\bullet \frac{\partial V}{\partial y} = 1^2 + 1 = 2 \Rightarrow E_y = -2$$

$$\bullet \frac{\partial V}{\partial z} = 3 \Rightarrow E_z = -3$$

Now calculate magnitude:

$$|\vec{E}| = \sqrt{(-6)^2 + (-2)^2 + (-3)^2}$$

$$= \sqrt{36 + 4 + 9} = \sqrt{49} = 7$$

Final Answer : (1)

35) **Explaining** : We are given:

- An **electric dipole** with **dipole moment** $\vec{p}$.
- Placed in a **uniform electric field** $\vec{E}$.
- Initially, the angle between $\vec{p}$ and $\vec{E}$ is **30°**, and it is **increased to 60°**,
- We are to calculate the **work done by the field** during this change in angle.

Concept : This question is based on

The **potential energy** U of a dipole in a uniform electric field is given by:

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

The **work done by the electric field** when the dipole rotates from angle θ_1 to θ_2 is:

$$W = U_{\text{initial}} - U_{\text{final}} = (-pE \cos \theta_1) - (-pE \cos \theta_2)$$

$$\Rightarrow W = pE (\cos \theta_2 - \cos \theta_1)$$

Solution : Given:

- $\theta_1 = 30^\circ$
- $\theta_2 = 60^\circ$

Substitute:

$$W = pE(\cos 60^\circ - \cos 30^\circ)$$

Use trigonometric values:

$$\bullet \cos 60^\circ = \frac{1}{2}$$

$$\bullet \cos 30^\circ = \frac{\sqrt{3}}{2}$$

So:

$$W = pE \left(\frac{1}{2} - \frac{\sqrt{3}}{2} \right) = pE \left(\frac{1 - \sqrt{3}}{2} \right)$$

Final Answer : (2)

36) **QUESTION EXPLANATION:**

You're given:

- A charge q is placed inside a **cavity** in a **neutral conductor**.
- An **external electric field** is now applied.

You are to determine what happens as a result.

CONCEPT:

□ Before the field is applied:

- A. A charge q inside the cavity **induces** a charge $-q$ on the **inner surface** of the cavity.
- B. To keep the conductor **neutral**, a charge $+q$ is induced on the **outer surface**.

□ When an external electric field is applied:

- A. **Electric field cannot exist inside the conductor's bulk**, so:
 - A. Charges **redistribute** on the **outer surface** to cancel the field inside.
 - B. However, the **inner surface charges remain unchanged** because the field from external sources does **not** penetrate into the cavity (due to electrostatic shielding).
- B. The **electric field inside the cavity** depends **only** on the enclosed charge (Gauss's law).
 - A. The **field inside the cavity remains unchanged**.
 - B. Hence, **no new force acts on the charge q** inside.

SOLUTION:

- A. The **inner surface charge distribution remains the same**.
- B. The **outer surface charges redistribute** to **cancel the external field** inside the conductor.

FINAL ANSWER:

(1) Only induced charge on outer surface will redistribute

37) **Explaining** : To find the flux coming out from the figure.

Concept : This question is based on Gauss law

Solution :

$$\phi = \frac{q}{12\epsilon_0}$$

To cover the charge we need 12 surface

Final Answer : (4)

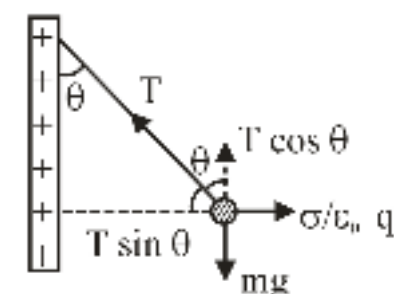
38) **Explaining** : To find the surface charge density of charged conducting sheet.

Concept : This question is based on

$$E = \frac{\sigma}{\epsilon_0}$$

- Field due to conducting sheet
- In equilibrium, net force on charged ball B is zero.

Solution :



$$\tan \theta = \frac{F_E}{mg} = \frac{q(\sigma)}{\epsilon_0 (mg)}$$

$$\sigma \propto \tan \theta$$

Final Answer : (3)

39) **Explaining** : To choose the correct relation between d , ℓ and θ

Concept : Projectile motion

Solution :

From the figure $a_x = 0$, $a_y = \frac{eE}{m}$

$\Rightarrow$ time of journey $t = \ell / u \cos \theta$

For upward motion up maximum height

$$t_{up} = \frac{t}{2} = \frac{\ell}{2u \cos \theta} \quad \text{and} \quad a_y = \frac{u_y}{t_{up}}$$

$$\text{For } d = \frac{u_y^2}{2a_y} = \frac{u^2}{2} \frac{\sin^2 \theta (\ell)}{(4 \sin \theta) (2u \cos \theta)}$$

$$4d = \ell (\tan \theta) \Rightarrow \tan \theta = \frac{4d}{\ell}$$

Final Answer : (2)

40) **Explaining** : To find the position of particle at $t = 1$ sec

Concept : Equipotential surface

Solution :

$$\text{From the figure } E = \frac{\Delta V}{\Delta r} = \frac{10}{0.1 \cos 45^\circ}$$

$E = 100\sqrt{2}$ v/m at 45° from x - axis

$$\Rightarrow \vec{E} = (100\hat{i} - 100\hat{j})$$

$$\vec{a} = (100\hat{i} - 100\hat{j}) \frac{q}{m}$$

$\Rightarrow$ position vector at $t = 1$ & $\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2$

$$\vec{r} = \frac{1}{2}(100)(\hat{i} - \hat{j}) \frac{q}{m} = \frac{50q}{m}(\hat{i} - \hat{j}) \text{ m.}$$

Final Answer : (2)

41) **Explaining** : To find electric field due to a short dipole

Concept : Electric field due to dipole

Solution :

$$E_1 = \frac{kp}{r^3} \sqrt{3\cos^2 30 + 1}$$

$$E_1 = \frac{kp}{r^3} \sqrt{\frac{9}{4} + 1} = \frac{k\rho}{2r^3} \sqrt{13}$$

$$\frac{kp}{r^3} = \frac{2}{\sqrt{13}} E_1$$

$$E_2 = \frac{kp}{r^3} \sqrt{3\cos^2 45 + 1}$$

$$E_2 = \frac{kp}{r^3} \sqrt{\frac{5}{2}}$$

$$E_2 = \frac{2}{\sqrt{13}} \sqrt{\frac{5}{2}} E_1 = \sqrt{\frac{10}{13}} E_1$$

Final Answer : (3)

42) **Explaining** : To find flux through conical angle

Concept : This question is based on gauss law

Solution :

From Gauss law,

$$\phi_{\text{hemisphere}} + \phi_{\text{Cone}} = \frac{q}{\epsilon_0} \dots\dots$$

$\phi \propto$ no of electric field lines passing through surface q is point charge which has uniformly distributed electric field lines thus half of electric field lines will pass through hemisphere & other half will pass through conical surface.

$$\frac{4q}{6\epsilon_0} = \frac{q}{2\epsilon_0}$$

$$n = 3$$

Final Answer : (1)

43) **Concept** : Multi-dimensional field gradients.

Solution:

$$U = -\vec{p} \cdot \vec{E}$$

$$U = pax - pby$$

$$\vec{F} = -\frac{\partial U}{\partial x} \hat{i} - \frac{\partial U}{\partial y} \hat{j}$$

$$\vec{F} = -(-pa)\hat{i} - (-pb)\hat{j}$$

The net force vector is $\vec{F} = pa\hat{i} + pb\hat{j}$.

Final Answer: Option (2)

44) **Explaining** : To calculate the work done

Concept : This question is based on potential energy

Solution :

Work done by external agent :

$$W_{\text{ext}} = \Delta U$$

$\Delta U \rightarrow$ Change in potential energy in taking the charge from initial to final configuration

$$\Rightarrow W_{\text{ext}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_f} - \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_i}$$

$$\text{Now, } r_f = \sqrt{(2-0)^2 + (2-0)^2 + (1-0)^2} = 3\text{m}$$

$$r_i = \sqrt{(4-0)^2 + (4-0)^2 + (2-0)^2} = 6\text{m}$$

$$\therefore W_{\text{ext}} = (9 \times 10^9) \times (10^{-8} \times 2 \times 10^{-6}) \left[\frac{1}{3} - \frac{1}{6} \right]$$

$$= 3 \times 10^{-5}$$

$$= 30 \times 10^{-6} \text{J}$$

Final Answer : (3)

45) **Explaining** : To select the current electric field pattern

Concept : This question is based on electric field lines of conductor

Solution :

Field lines must be normal to the surface of conductor

Final Answer : (1)

CHEMISTRY

46)

Explaining : Energy spectrum.

Concept : This question is based on energy spectrum.

Solution :

$$\text{Number of emission line} = \frac{(5-1)(5-1+1)}{2} = 10$$

Final Answer : (3)

47) **Explaining :** Bohr's atomic model.

Concept : This question is based on Bohr's atomic model radius.

Solution :

$$r \propto n^2 \Rightarrow \frac{r_1}{r_2} = \frac{(2)^2}{(3)^2} \Rightarrow \frac{R}{r_2} = \frac{4}{9}$$
$$\Rightarrow r_2 = \frac{9R}{4} = 2.25 R$$

Final Answer : (4)

48) **Explaining :** Energy of e^- .

Concept : This question is based on Bohr's atomic model.

Solution :

$(n_1 = 1) \quad (n_2 = 3) \quad (2\text{nd excited state})$

$$\Delta E = 13.6 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Final Answer : (4)

49) **Explaining :** Isodiapher

Concept : This question is based on isodiapher.

Solution : $(n - p) \rightarrow \text{Same}$

Final Answer : (2)

50) **Explaining :** Isoelectronic species.

Concept : This question is based on isoelectronic species.

Solution : (Isoelectronic Species $\rightarrow$ same number of electron)

Final Answer : (1)

51) **Explaining :** Atomic parameters.

Concept : This question is based on atomic parameters.

Solution : (z = number of proton $\rightarrow$ doesn't change with charge)

Final Answer : (1)

52) **Explaining :** Atomic parameters.

Concept : This question is based on atomic parameters.

Solution : Electron is not present in nucleus.

Final Answer : (4)

53) **Explaining :** Atomic parameters.

Concept : This question is based on atomic parameters.

Solution :

$$m_n = 1.008 \text{ amu} ; m_p = 1.007 \text{ amu}$$

$$\frac{m_n}{m_p} > 1$$

Final Answer : (2)

54) **Explaining :** Rutherford's experiment.

Concept : This question is based on Rutherford's experiment.

Solution : (According to theory)

Final Answer : (3)

55)

Explaining : Rutherford's α -particle scattering experiment ?

Concept : This question is based on Rutherford's α -particle scattering experiment ?

Solution : The positive charge is concentrated in a very small volume of the atom.

Final Answer : (1)

56)

Explaining : Angular momentum.

Concept : This question is based on Bohr's atomic model.

$$\text{Solution : } p\text{-shell} \rightarrow n = 6 \Rightarrow mvr = \frac{6h}{2\pi} = \frac{3h}{\pi}$$

Final Answer : (1)

57)

Explaining : Atomic parameters.

Concept : This question is based on atomic parameters.

Solution :

$$A = 39 \quad (n_n = n_p + 1)$$

$$n_n + n_p = 39$$

$$n_p + 1 + n_p = 39$$

$$\Rightarrow n_p = 19 \Rightarrow n_e = 19$$

$$n_n = 20$$

Final Answer : (1)

58)

Explaining : Energy of e^- .

Concept : This question is based on Bohr's atomic model.

$$\text{Solution : } E = -13.6 \times \frac{z^2}{n^2} = -13.6 \times \frac{(1)^2}{(2)^2} = -3.4 \text{ eV}$$

Final Answer : (2)

59)

Explaining : Kinetic energy.

Concept : This question is based on kinetic energy.

Solution : as $r \uparrow$ K. $E \downarrow \Rightarrow x > y$

Final Answer : (2)

60) **Explaining :** Spectrum.

Concept : This question is based on spectrum.

Solution :

$$\frac{(\nu_{\max})_{\text{Lyman}}}{(\nu_{\max})_{\text{Brackett}}} = \frac{\left[\frac{1}{(1)^2} - \frac{1}{\infty} \right]}{\left[\frac{1}{(4)^2} - \frac{1}{\infty} \right]}$$

$$= \frac{16}{1} \Rightarrow 16 : 1$$

Final Answer : (2)

61)

Explaining : No. of moles.

Concept : This question is based on No. of moles.

$$\text{Solution : } n_{\text{O}_3} = \frac{12.044 \times 10^{22}}{N_A} = 0.2 \text{ mol}$$

Final Answer : (4)

62)

Explaining : Mole fraction.

Concept : This question is based on mole fraction.

Solution :

$$n_A = 5 \text{ mol}$$

$$n_B = 15 \text{ mol}$$

$$\Rightarrow x_A = \frac{5}{5+15} = \frac{5}{20} = \frac{1}{4} = 0.25$$

Final Answer : (3)

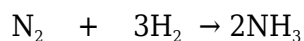
63)

Explaining : Stoichiometry.

Concept : This question is based on stoichiometry.

Solution :

1L 3L



3L $\rightarrow 3 \times 3 = 9\text{L}$ of H_2

Final Answer : (3)

64)

Explaining : Stoichiometry

Concept : This question is based on stoichiometry.

Solution :

(Concept of L.R)

ratio A is in excess $\rightarrow \frac{5}{2} < \frac{3}{6} \rightarrow \text{B is L.R}$

Final Answer : (1)

65) **Explaining :** Percentage of atom in a compound.

Concept : This question is based on percentage of atom in a compound.

Solution : % of H = $\frac{4}{16} \times 100 = 25\%$

Final Answer : (1)

66)

Explaining : Empirical formula.

Concept : This question is based on empirical formula.

Solution :

$$\frac{\text{A}}{9 \times 10^{20}} : \frac{\text{B}}{3 \times 10^{20}}$$

$\Rightarrow 3 : 1 \Rightarrow \text{A}_3\text{B}$

Final Answer : (2)

67)

Explaining : Molality.

Concept : This question is based on molality.

Solution :

$$m = \frac{(\text{moles})_{\text{solute}}}{(W_{\text{solvent}})_{\text{kg}}}$$
$$= \frac{2}{20} = 0.1 \text{ m}$$

Final Answer : (1)

68)

Explaining : Energy of photon.

Concept : This question is based on Bohr's electronic model.

Solution : $\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = 2$

Final Answer : (2)

69)

Explaining : Angular momentum.

Concept : This question is based on Bohr's theory.

Solution : $L = \frac{nh}{2\pi} = \frac{5h}{2\pi} = 2.5 \frac{h}{\pi}$

Final Answer : (4)

70)

Explaining : Spectrum

Concept : This question is based on Bohr's theory.

Solution : $\frac{n(n-1)}{2} = 15 \Rightarrow n = 6; \frac{1}{\lambda} = R_H \left[\frac{1}{1^2} - \frac{1}{6^2} \right] \Rightarrow \lambda = \frac{36}{35R_H}$

Final Answer : (2)

71) **Explaining :** Mole parameters.

Concept : This question is based on mole parameters.

Solution : $\frac{2}{44} : \frac{5}{44} \Rightarrow 2 : 5$

Final Answer : (2)

72)

Explaining : Molecular formula.

Concept : This question is based on molecular formula.

Solution :

$\frac{120}{30} = 4 = n^0$

molecular formula = $n_0 \times$ (Empirical formula)

$= 4 \times \text{CH}_2\text{O} = \text{C}_4\text{H}_8\text{O}_4$

Final Answer : (2)

73)

Explaining : Velocity of electron.

Concept : This question is based on Bohr's model.

Solution : $v \propto \frac{Z}{n}$

Final Answer : (1)

74)

Explaining : Vapour density

Concept : This question is based on vapour density.

Solution :
$$\text{V.D.} = \frac{\text{molar mass}}{2}$$

Final Answer : (4)

75) **Explaining :** % of a compound.

Concept : This question is based on % of a compound.

Solution :

$$\text{mass \% of Na} = \frac{1 \times \text{At.wt}}{M_w} \times 100$$
$$\frac{1 \times 23}{M_w} \times 100$$

$$11.5 = \frac{M_w}{100} \times 100$$

$$M_w = 200 \text{ amu}$$

Final Answer : (2)

76) **Explaining :** Heisenberg's principle.

Concept : This question is based on Heisenberg's principle.

Solution :
$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

Final Answer : (2)

77) **Explaining :** Mole

Concept : This question is based on mole.

Solution : $(\text{NH}_4)_2 \text{Cr}_2\text{O}_7 \rightarrow 19 \text{ atoms}$ 1 mol contains = $19 \times N_A$ (19 mol atoms)

Final Answer : (3)

78) **Explaining :** Angular momentum.

Concept : This question is based on Bohr's model.

Solution : Difference =
$$\frac{3h}{2\pi} - \frac{h}{2\pi} = \frac{2h}{2\pi} = \frac{h}{\pi}$$

Final Answer : (1)

79)

Explaining : Radius of shells.

Concept : This question is based on Bohr's model.

Solution :
$$\frac{r_1}{r_2} = \frac{1}{9} = \frac{n_1^2}{n_2^2} = \frac{1}{9} \Rightarrow \frac{n_1}{n_2} = \frac{1}{3} \rightarrow K \text{ \& M}$$

Final Answer : (3)

80)

Explaining : Wavelength

Concept : This question is based on wavelength.

Solution :

$$\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$
$$\Rightarrow \frac{1}{300} = \frac{1}{600} + \frac{1}{\lambda_2} \Rightarrow \lambda_2 = 600 \text{ nm}$$

Final Answer : (3)

81)

Explain Question : We need to arrange particles by their specific charge (charge/mass ratio, $\frac{q}{m}$).

Concept : Specific charge = $\frac{q}{m}$

Solution :

A. Electron: $\frac{q}{m} \approx 1.758 \times 10^{11} \text{ C/kg}$

B. Proton: $\frac{q}{m} \approx 9.58 \times 10^7 \text{ C/kg}$

C. Neutron: $\frac{q}{m} = 0$ (neutral)

D. Na^+ : $q/m \approx 9.58 \times 10^7 / 23 \approx 4.16 \times 10^6 \text{ C/kg}$

Order : $e^- > p > \text{Na}^+ > n$

Final Answer : option (2)

82) **Explaining :** The angular momentum.

Concept : This question is based on Bohr's theory

Solution :

$$\text{Angular momentum (mvr)} = \frac{nh}{2\pi}$$
$$= 4 \times \frac{h}{2\pi} = 2 \frac{h}{\pi}$$

Final Answer : (2)

83) **Explaining :** Velocity of the electron.

Concept : This question is based on Bohr's theory.

$$\text{Solution : } (v \propto \frac{1}{n} \text{ (as } z = \text{same)}) \Rightarrow \frac{v_1}{v_2} = \frac{n_2}{n_1} = \frac{5}{3}$$

Final Answer : (2)

84) **Explaining** : Ionization energy.

Concept : This question is based on Bohr's theory.

Solution : $I.E = -E_1 = x \Rightarrow E_1 = -x$

Final Answer : (3)

85) **Explaining** : Emission line in H-atom.

Concept : This question is based on spectrum.

Solution : $\bar{\nu} = R(1)^2 \left[\frac{1}{(2)^2} - \frac{1}{(3)^2} \right] = \frac{5R}{36}$

Final Answer : (1)

86) **Explaining** : No. of moles.

Concept : This question is based on mole concept.

Solution : $n_{c-atom} = \frac{12}{12} = 1 \text{ mol} \xrightarrow{\times 6} 6 \text{ mol } e^-$

Final Answer : (1)

87) **Explaining** : Moles of atom.

Concept : This question is based on mole concept.

Solution :

$n_{O_2} = \frac{32}{32} = 1 \text{ mol of } O_2 \xrightarrow[\text{(atomicity)}]{\times 2} 2 \text{ mol of O-atoms}$

Final Answer : (1)

88) **Explaining** : Combustion reaction.

Concept : This question is based on stoichiometry.

Solution :

$$\underset{3}{C_2H_6} + \underset{7}{2} O_2 \rightarrow 2CO_2 + 3H_2O$$

moles $\frac{3}{30} = 0.1 \text{ mol} \rightarrow 0.2 \text{ mol}$
 $CO_2 \Rightarrow 4.48 \text{ L of } CO_2 \text{ at STP } ((V_L)_{STP} = n \times 22.4)$

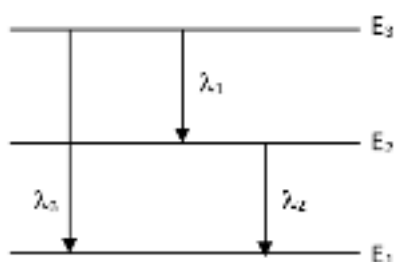
Final Answer : (3)

89)

Explaining : Energy of e^- .

Concept : This question is based on energy of e^- .

Solution :



$$\begin{aligned} E_{3-1} &= E_{3-2} + E_{2-1} \\ \frac{E_{3-1}}{hc} &= \frac{E_{3-2}}{hc} + \frac{E_{2-1}}{hc} \Rightarrow \frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} \\ h\nu_3 &= h\nu_1 + h\nu_2 \\ \nu_3 &= \nu_1 + \nu_2 \end{aligned}$$

Final Answer : (3)

90)

Explaining : Ionisation energy.

Concept : This question is based on Bohr's atomic model.

Solution :

$$IE = -E_1$$

$$E_1 = -13.6 \times Z^2 \quad n = 1$$

$$E_{Li+2} = -13.6 \times (3)^2 \quad Z = 3$$

$$E_{Be+3} = -13.6 \times (4)^2 \quad Z = 4$$

$$\frac{I.E. (Li^{+2})}{I.E. (Be^{+3})} = \frac{9}{16}$$

Final Answer : (3)

BIOLOGY

91) **Solution :** Given animal is *Nereis* (Phylum - Annelida)

Final Answer : (3)

92) **Solution :** 1-d, 2-c, 3-a, 4-b

Final Answer : (3)

93) **Solution :**

- A. **Bilateral symmetry** is seen in animals where body can be divided into equal halves only in one plane.
- B. *Ascaris* (**Nematoda**), *Nereis* (**Annelida**), and Scorpion (**Arthropoda**) all show **bilateral symmetry**.
- C. Other options include animals with **radial or asymmetrical** body plans.

Final Answer : (3)

94) **Solution :** Pneumatic bones, dry skin, crop & gizzard, warm-blooded → Found in **birds**.

Final Answer : (2)

95) **Solution:**

A - Coelomate - Cockroach

B - Pseudocoelomate - *Ascaris*

C - Acoelomate - Tapeworm

Final Answer: (2)

96) **Solution** : a = Adamsia - Polyp form; b = Aurelia - Medusa form

Final Answer : (3)

97) **Solution**: Marine, radially symmetrical, diploblastic, tissue-level, extracellular & intracellular digestion, bisexual, external fertilization, only sexual reproduction →

Ctenophora.

Final Answer: (3)

98) **Solution** : Only parasitic flatworms have hooks and suckers; not all.

Final Answer : (1)

99) **Solution** : Molluscs are unsegmented animals.

Their body is divided functionally into head, muscular foot, and visceral hump, but these are not true segments.

Final Answer : (1)

100) **Solution** :

Saccoglossus → Hemichordata

Cuttle fish, Sea hare, Chiton → Mollusca

Culex → Arthropoda Therefore, only 5 animals belong to Echinodermata.

Final Answer : (4)

101)

Solution :

Proboscis Gland: It is the excretory organ found in Hemichordates.

Location: It is situated in the proboscis (front part) of the body.

Final Answer : (4)

102) **Solution**:

A - Agnatha - Lacks jaw

B - Gnathostomata - Bears jaw

C - Tetrapoda - Limbed vertebrates

D - Osteichthyes - Bony fishes **Final answer**: (4)

103) **Solution** : The given animal is Scorpion (Class - Arachnida, Phylum - Arthropoda)

Excretory organ - Coxal gland

Respiratory organ - Book lungs

Final Answer : (3)

104)

Solution : Aschelminthes (Nematoda) have the first complete alimentary canal with a muscular pharynx.

Others like Platyhelminthes, Ctenophora, Coelenterata have incomplete gut.

Final Answer : (2)

105)

Solution :

Air bladder – Chondrichthyes (Scoliodon) have no swim bladder, Osteichthyes (Hippocampus) have a swim bladder for buoyancy.

Terminal mouth – Chondrichthyes usually have ventral mouth, Osteichthyes often have terminal mouth.

Oviparous – Many Chondrichthyes are viviparous, whereas Osteichthyes are generally oviparous.

Final Answer : (2)

106) **Solution :** Protochordates are primitive chordates.

Include Tunicata (sea squirts) and Cephalochordata (lancelets).

They show chordate features but are simpler than vertebrates.

Final Answer : (3)

107) **Solution :**

Egg-laying mammal → Platypus

Flying mammal → Pteropus

Reason: Just as Platypus is a unique mammal laying eggs, Pteropus is a mammal capable of flight.

Final Answer: (4)

108) **Solution :**

Macropus → Mammal (kangaroo) → hair, mammary glands

Pteropus → Mammal (bat) → flying mammal

Struthio → Bird (ostrich) → feathers, lays eggs

Final Answer : (2)

109) **Solution :** Correct Match: Salamandra belongs to Class Amphibia. It has a tympanum (ear drum) and lacks external ears.

Incorrect Options:

- A. **Aurelia:** Has tissue level organisation, not organ level.
- B. **Ascaris:** Belongs to Aschelminthes, not Annelida; its body is unsegmented.
- C. **Pteropus:** (Flying fox/Bat) is a viviparous mammal, not oviparous (egg-laying).

Final Answer : (3)

110) **Solution :** Amphibians live in **both aquatic and terrestrial habitats**, not only terrestrial.

Final Answer : (4)

111) **Solution :** Peculiar character of Porifera: presence of choanocytes (collar cells) that line the spongocoel and help in filter feeding.

Other features like diploblastic, habitat, and intracellular digestion are not unique to Porifera.

Final Answer : (3)

112) Solution :

C → Protochordates are not vertebrates.

D → Nerve cord in chordates is dorsal, not ventral.

Final Answer : (2)

113) Solution : Kingdom Animalia members are:

- A. Heterotrophic & multicellular → cannot make their own food
- B. Eukaryotic, no cell wall → flexible body cells
- C. Depend on plants → directly (herbivores) or indirectly (carnivores)

Hence, all statements are correct.

Final Answer : (4)

114) Solution :

- A. Sea horse & Flying fish → both poikilothermic (cold-blooded)
- B. *Ascaris* & *Ancylostoma* → unsegmented (not metameric)
- C. *Pteropus* & *Ornithorhynchus* → *Ornithorhynchus* is oviparous, not viviparous
- D. Garden lizard & Crocodile → lizard has 3-chambered heart, crocodile 4-chambered

Final Answer : (2)

115) Solution :

- 1. Not all sponges are marine (some are freshwater).
- 2. Not all mammals are viviparous (e.g., platypus).
- 3. Crocodile has four-chambered heart, not three.
- 4. All bony fishes (Osteichthyes) have gills covered by operculum.

Final Answer : (4)

116) Solution :

Poikilothermic + dry skin → characteristic of reptiles

Chameleon is a reptile → cold-blooded + dry, scaly skin

Others:

Salamandra → moist skin (amphibian)

Hippocampus → fish (not dry skin)

Ornithorhynchus → mammal (warm-blooded)

Final Answer : (3)

117) Solution :

- (a) Phylum → Chordata (S)
- (b) Subphylum → Vertebrata (T)
- (c) Division → Tetrapoda (R)
- (d) Super class → Agnatha (P)
- (e) Class → Cyclostomata (Q)

Final Answer : (3)

118) Solution :

Birds & mammals have a 4-chambered heart → complete separation of oxygenated &

deoxygenated blood.

They show double circulation → pulmonary + systemic.

Final Answer : (2)

119) **Solution :** A-iv, B-iii, C-i, D-ii

Mollusca → Radula (A-iv)

Arthropoda → Malpighian tubules (B-iii)

Hemichordata → Proboscis gland (C-i)

Echinodermata → Radial symmetry (D-ii)

Final Answer : (4)

120) **Solution :**

Chelone (turtle) → 3-chambered heart;

Scoliodon & Hippocampus → 2-chambered;

Pteropus → 4-chambered.

Final Answer : (3)

121) **Explanation:** Binomial nomenclature was given by Linnaeus, not Ernst Mayr.

Final Answer : (3)

122) **Explanation:** Man - *Homo sapiens* , Mango - *Mangifera indica* , Housefly - *Musca domestica*

Final Answer : (1)

123) **Explanation:** Only the genus is capitalized, and names are italicized or underlined when handwritten.

Final Answer : (2)

124) **Explanation:** A taxon is any rank or level in taxonomic classification.

Final Answer : (2)

125) **Explanation:** Maximum number of species present in kingdom.

Final Answer : (4)

126) **Explanation:** *Mangifera - indica*, *Triticum - aestivum*

Final Answer : (4)

127) **Explanation:** Animals belonging to the Order Carnivora are meat-eaters.

Dog, Tiger → Carnivorous animals

Cow, Goat → Herbivorous animals

Only the dog and the tiger belong to the Order Carnivora.

Final Answer : (1)

128) **Explanation:**

- In the classification of animals → the term 'Phylum' is used.
- In the classification of plants → the term 'Division' is used.

Final Answer : (4)

129) **Explanation:** Statement III is incorrect—scientific names must be underlined when handwritten.

Final Answer : (1)

130) **Explanation:** IV is incorrect—*Mangifera indica* belongs to the family Anacardiaceae

Final Answer : (1)

131) **Explaining :** Lichens are symbiotic association between algae and fungi which are pollution indicators and not place in any kingdom by R. H. Whittaker.

Final Answer : (4)

132) **Explaining :** Mycorrhiza is indeed a symbiotic association between the roots of higher plants and fungi. This is a correct statement. W.M. Stanley crystallized Tobacco Mosaic Virus (TMV), proving that viruses could be crystallized. This statement is correct.

Final Answer : (2)

133) **Explaining :** A-III, B-II, C-IV, D-I

Rhizopus — Bread mould; *Ustilago* — Smut fungus

Puccinia — Rust fungus; *Agaricus* — Mushroom

Final Answer : (1)

134) **Explaining :** A-IV, B-III, C-II, D-I

Euglenoid — Switching between photosynthetic and heterotrophic mode

Dinoflagellate — Photosynthetic

Slime mould — Saprophytic

Plasmodium — Parasitic

Final Answer : (3)

135) **Explaining :** All characters found in ascomycetes.

Final Answer : (1)

136) **Explaining :** Euglenoids have flexible body due to covered by protein rich layer called pellicle.

Final Answer : (1)

137) **Explaining :** A - (iv), B - (iii), C - (i), D - (ii)

Phycomycetes — Algal fungi

Ascomycetes — Sac fungi

Basidiomycetes — Club fungi

Deuteromycetes — Fungi imperfecti

Final Answer : (3)

138) **Explaining** : A - Cell wall, B - Cell membrane, C - DNA, D - Heterocyst, E - Mucilagenous sheath

Final Answer : (3)

139) **Explaining** : Bacteria, blue green algae, fungus Mosses and fern, Gymnosperms and angiosperm they have cell wall.

Final Answer : (4)

140) **Solution** : According to R.H. Whittaker, *Chlamydomonas* & *Chlorella* are placed in Kingdom Protista.

Final Answer : (3)

141) **Solution** : A unicellular eukaryotic wall less microbe was collected by a student from rain water. His teacher grouped the organism in **animalia** kingdom according to Linnaeus system and **protista** kingdom according to Whittaker's system.

Final Answer : (4)

142) **Solution** : Cellulosic cell wall is absent in most of the fungi, Euglenoids & Bacteria

Final Answer : (1)

143) **Solution** : Archaeobacteria differ from other bacteria in having different cell wall and cell membrane.

Final Answer : (3)

144) **Solution** : Beside paddy fields cyanobacteria (BGA) are also found inside vegetative part of *cycas*.

Final Answer : (3)

145) **Solution**: Characteristics of Cyanobacteria:

Nitrogen fixation

Peptidoglycan cell wall

Non-oxygenic photosynthesis → Incorrect

Antibiotic production → No

Final Answer: (2)

146) **Solution** :

(A) Correct - Mycoplasma lacks a cell wall.

(B) Correct - It is referred to as the "Joker of the Plant Kingdom."

(C) Incorrect - It is sensitive to tetracycline, not resistant.

(D) Correct - It exhibits osmotrophic nutrition.

(E) Correct - The cell membrane is trilaminar.

(F) Correct - It is a facultative parasite.

Final Answer : (3)

147) **Solution:**

They are saprophytic.

They form a plasmodium.

Spores are dispersed by air.

Final Answer: (3)

148) **Solution :** A-i, B-iii, C-iv, D-ii

(A)	Production of methane	(i)	Methanogen bacteria
(B)	Most extensive metabolic diversity	(iii)	Monera
(C)	Hot spring	(iv)	Thermoacidophiles
(D)	Extreme salty area	(ii)	Halophiles

Final Answer : (1)

149) **Explain Question :** The question asks to match the classes of fungi (Column-I) with their respective examples (Column-II)

Concept : This question is based on classes of fungi

Solution :

A. Ascomycetes : Matches with (iii) *Claviceps*. *Claviceps* is a genus of fungi in the family Clavicipitaceae, belonging to the Ascomycetes.

B. Deuteromycetes : Matches with (i) *Trichoderma*. *Trichoderma* is a genus of fungi in the family Hypocreaceae, belonging to the Deuteromycetes (also known as Fungi Imperfecti).

C. Phycomycetes : Matches with (ii) *Albugo*. *Albugo* is a genus of plant-parasitic oomycetes, which were formerly classified under Phycomycetes.

D. Basidiomycetes : Matches with (iv) *Agaricus*. *Agaricus* is a genus of mushrooms containing both edible and poisonous species, belonging to the Basidiomycetes.

Final Answer : (3)

150) **Explain Question:** Which fungi cause rust and smut diseases?

Concept: This question is based on Basidiomycetes

Solution :

Puccinia → rust fungus (e.g., *Puccinia graminis* causes wheat rust).

Ustilago → smut fungus (e.g., *Ustilago maydis* causes corn smut).

Final answer : 2

151) **Explain Question :** The question asks which spores in fungi are involved in sexual reproduction.

Concept : This question is based on sexual reproduction in fungi.

Solution : From the options, the spores linked to sexual reproduction are Oospore, Ascospore, and Basidiospore.

Final answer : (4)

152) **Solution :** "Late blight of potato" Disease is caused by a member of Phycomycetes.

Final Answer : (1)

153) **Explain Question:** The question asks the name of the person who demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants.

Concept: Virus, Viroid and Prions.

Solution: Beijerinck demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants.

Final answer: (2)

154) **Question Explanation:** The question is asking about the causative agent of CJD.

Concept: Prions

Solution: Creutzfeldt-Jakob disease (CJD) is a fatal neurodegenerative disorder in humans. It is caused by prions, which are abnormally folded, proteinaceous infectious particles.

Final Answer: (1)

155) **Solution :** Prions consist of abnormally folded protein.

Final Answer : (4)

156) **Explanation:** These are all genera of green algae. *Ulothrix* and *Spirogyra* are filamentous green algae, and *Chara* is a complex branched green alga.

Answer: (2) Chlorophyceae

157) **Explanation:** Zoospores are asexual reproductive units in algae that possess flagella, allowing them to move in water. Akinetes, aplanospores, and hypnospores are non-motile.

Answer: (1) Zoospore

158) **Explanation:** Algae are classified primarily based on their types of pigments, the chemical nature of their stored food, and the composition of their cell walls.

Answer: (4) Place of occurrence

159) **Explanation:** Most algae have unicellular sex organs. *Chara* is an exception in the plant kingdom as it has complex, multicellular, and jacketed sex organs (the globule and nucule).

Answer: (1) Chara

160) **Explanation:** *Chlamydomonas* is a unicellular green alga that typically contains a single, large cup-shaped chloroplast and stores energy as starch.

D = *Chlamydomonas*, Starch, Cup-shaped

Answer: (1)

161) **Explanation:** Gemmae are specialized asexual buds produced in small receptacles called gemma cups on the dorsal surface of the liverwort *Marchantia*.

Answer: (2) Marchantia thallus

162) **Explanation:** *Marchantia* is dioecious (separate male and female plants). The female

reproductive organs (archegonia) are borne on a stalked structure called the archegoniophore.

Answer: (1) *Marchantia*

163) **Explanation:** The diagram shows a male *Marchantia* plant. "B" is the stalk carrying the male organs (Antheridiophore), "C" is the gemma cup for asexual reproduction, and "D" are the root-like rhizoids.

Answer: (2) A-Male Thallus, B-Antheridiophore, C-Gemma cup, D-Rhizoids

164) **Explanation:** In liverworts, spores germinate directly into the adult plant body. The "protonema" stage is a characteristic specific to mosses.

Answer: (1) Liverworts

165) **Explanation:** The first stage of the gametophyte in mosses, which develops directly from a spore, is the filamentous or creeping protonema.

Answer: (3) Mosses

166) **Explanation:** A moss sporophyte consists of a foot, seta, and capsule. Elaters (cells that aid in spore dispersal) are found in liverworts like *Marchantia* but are absent in mosses.

Answer: (4) Elaters

167) **Explanation:** In the moss *Funaria*, the spore germinates into a primary protonema. Buds then develop on the protonema, which grow into the upright leafy gametophore.

Answer: (4) First protonema, then forms a bud and then a leafy gametophyte

168) **Explanation:** In mosses, the sporophyte (C) is composed of the Capsule (A) and Seta (B). It remains attached to and dependent on the leafy Gametophyte (D).

Answer: (2) A-Capsule, B-Seta, C-Sporophyte, D-Gametophyte

169) **Explanation:** *Equisetum* (Horsetail) produces only one type of spore (homosporous). *Selaginella* and *Salvinia* are heterosporous (produce two types of spores).

Answer: (2) *Equisetum*

170) **Explanation:** The prothallus is the inconspicuous, small, multicellular, free-living, photosynthetic gametophyte found in pteridophytes like *Pteris* (fern).

Answer: (4) *Pteris*

171) **Explanation:** Ferns produce large, spirally coiled male gametes with many flagella to swim through water to reach the egg.

Answer: (1) Ferns

172) **Explanation:** *Sphagnum* (Peat moss) accumulates over time in wetlands to form peat, which is used as fuel and for its water-retention properties. **Answer: (4) *Sphagnum***

173) **Explanation:** When a haploid spore falls on moist soil, it germinates to form the heart-shaped gametophyte known as the prothallus.

Answer: (4) From spore

174) **Explanation:** *Cycas* is a gymnosperm. Like most gymnosperms, its xylem contains tracheids but lacks true vessels (vessels are a characteristic of angiosperms).

Answer: (1) Xylem have vessels

175) **Explanation:** The primary reason *Cycas* is classified as a gymnosperm is that its ovules are not enclosed in an ovary; they are "naked" on the surface of megasporophylls.

Answer: (1) Naked ovules

176) **Explanation:** (C) and (D) are correct. (A) is incorrect because gymnosperms *have* xerophytic traits. (B) is incorrect because coralloid roots host cyanobacteria, not mycorrhiza (mycorrhiza are in *Pinus*).

Answer: (1) 2 correct, 2 incorrect

177) **Explanation:** Since there is no ovary in gymnosperms, no fruit is formed. Therefore, the seeds are exposed or "naked."

Answer: (1)

178) **Explanation:** This is false. Many gymnosperms like *Cycas* are dioecious (male and female cones on different plants).

Answer: (4) All Gymnosperms are Monoecious

179) **Explanation:** In gymnosperms, the megaspore is never released; it remains inside the nucellus (megasporangium) to develop into the female gametophyte.

Answer: (2) Inside the megasporangium

180) **Explanation:** Statement (D) is incorrect. Pteridophytes are the first terrestrial vascular plants, but they do **not** produce seeds; they reproduce via spores.

Answer: (4) One